Number Theory

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Contents

1 Divisibility, Factorization, and Modular Arithmetic

§1.1 Divisibility, Euclidean Algorithm, and Factorization

For convention, we note that $\mathbb{N} = \{0, 1, \dots\}$.

Definition 1.1.1. For $a, b \in \mathbb{N}$, we say that a **divides** b, writing $a \mid b$, if $\exists n \in \mathbb{N}$ such that $an = b$.

Remark 1.1.2 — Note that for $a, b, c \in \mathbb{N}$, if $a \mid b, a \mid c$, then $a \mid bn + cm$ for any integers n, m.

Remark 1.1.3 — $\mathbb{N}$ with $\mid$, divisibility, is a partially ordered set:

- $a \mid a$ (reflexivity)
- $a \mid b, b \mid a \implies a = b$
- $a \mid b, b \mid c \implies a \mid c$ (transitivity)

Definition 1.1.4. $p \in \mathbb{N}$ is **prime** is $a \mid p \implies a = 1$ or p.

We can extend the above for $\mathbb{Z} = \mathbb{N} \cup -\mathbb{N}$. In this context, we define primes as numbers such that $a \mid p \implies a = \pm 1$ or $\pm p$.

Example 1.1.5

$\mathbb{Z}^\times = \{\pm 1\}$.

Definition 1.1.6. We define the **Gaussian integers** as elements of the set $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$.

Definition 1.1.7. The complex norm of $z \in \mathbb{Z}[i]$ is defined as $z \cdot \bar{z}$. We denote this as $N(z)$.

Remark 1.1.8 — The complex norm is multiplicative. This implies that for $a, b \in \mathbb{Z}[i]$, we have $a \mid b \implies N(a) \mid N(b)$.

Lemma 1.1.9

$\mathbb{Z}[i]^\times = \{\pm 1, \pm i\}$.

Proof. We use the complex norm here. For $a \in \mathbb{Z}[i]$, so $N(a) = 1$, implying that if $a = x + iy$, then $x^2 + y^2 = 1$. This gives the desired cases. $\square$

§1.1.i The Euclidean Algorithm

Definition 1.1.10. The **greatest common divisor(g.c.d)** of $a, b \in \mathbb{N}$, (a, b), is the largest positive integer such that $(a, b) \mid a$ and $(a, b) \mid b$.

We can extend the above for $\mathbb{Z}$ again to attain a definition for the greatest common divisor of two integers.

Remark 1.1.11 — Note that for $a, b \in \mathbb{Z}$, if $a = 0$, then $(a, b) = b$.

Lemma 1.1.12 (Euclidean Algorithm)
Let $a, b \in \mathbb{Z}$, and $b > 0$. Then,
$$a = bq + r,$$
such that $q \in \mathbb{Z}$, $0 \leq r < |b|$.

Proof. We have that
$$\frac{a}{b} = q + \frac{r}{b},$$
with $0 \leq r < |b|$. Then, we take q to be the greatest integer that is less that $\frac{a}{b}$, so $|\frac{r}{b}| < 1$. □

Lemma 1.1.13
Let $a, b \in \mathbb{Z}$. Then, for any integer n, we have $(a, b) = (b, a - nb)$.

Proof. From the definition of (a, b) it suffices to show that any positive integer divides m both a and b if and only if it divides both b and $a - nb$. But if m divides a and b, it clearly divides $a - nb$, and if it divides b and $a - nb$, it clearly divides a. □

Definition 1.1.14. In an integral domain R, let $a, b \in R$. Then, a and b are **associated**, $a \sim b$, if $a \mid b, b \mid a$.

Remark 1.1.15 — $\sim$ is an equivalence relation.

Lemma 1.1.16
Given an integral domain R, $a \sim b \iff a = ub$, for $u \in R^\times$.

We introduce another notion for the g.c.d. here.

Definition 1.1.17. Let R be an integral domain, and $a, b \neq 0$. We say that a $d \in R$ is a **greatest common divisor** for a, b if

1. $d \mid a, d \mid b$
2. For $d' \in R$ such that $d' \mid a, d' \mid b \implies d' \mid d$.

Remark 1.1.18 — Note that if the g.c.d. of a, b in an integral domain R exists, it is unique up to associates.

Theorem 1.1.19

Given $a, b \in \mathbb{Z}$, there exists $x, y \in \mathbb{Z}$ such that

$$(a, b) = xa + yb.$$

Proof. This follows by the Euclidean Algorithm. □

§1.1.ii Unique Factorization

Definition 1.1.20. $x, y \in \mathbb{Z}$ are relatively prime if $(x, y) = 1$.

Lemma 1.1.21

Given $a, b, n \in \mathbb{Z}$, $(n, a) = 1$, we have $n \mid ab \implies n \mid b$.

Corollary 1.1.22

Let $p \in \mathbb{Z}$ be prime, and $a, b \in \mathbb{Z}$. Then, $p \mid a$ or $p \mid b$.

Theorem 1.1.23 (Fundamental Theorem of Arithmetic)

Let n be a positive integer, and suppose we have two factorizations:

$$n = p_1 \dots p_r = q_1 \dots q_s.$$

with the p_i and q_j prime (and positive). Then $r = s$ and the p_i are a rearrangement of the q_j.

Proof. Suppose otherwise, and let n be the smallest positive integer where this does not hold. We then have $p_1 \mid q_1 \dots q_s$, so either $p_1 \mid q_1$ or $p_q \mid q_2 \dots q_s$. Proceeding inductively, we have $p_1 \mid q_i$ for some i; since q_i is prime this means $p_i = q_i$. We then have:

$$p_2 \dots p_r = q_1 \dots q_{i-1} q_{i+1} \dots q_s.$$

This product is smaller than n, so we must have that $r - 1 = s - 1$ and the p's (except p_q) are a rearrangement of the q's (except q_i). But since $p_1 = q_i$, we have $r = 2$ and the p's are a rearrangement of the q's. □

§1.2 Linear Diophantine Equations

Definition 1.2.1. A **Diophantine equation** is a system of polynomial equations with integer coefficients where we seek integer solutions.

Here are examples of Diophantine equations.

Theorem 1.2.2 (Four Square Theorem)
Any positive integers can be written as a sum of four squares. That is,

$$n = a^2 + b^2 + c^2 + d^2, \quad a, b, c, d \in \mathbb{Z}.$$

Theorem 1.2.3 (Two Square Theorem)
There is a necessary and sufficient condition for $n \in \mathbb{N}$ to be a sum of two squares in terms of its prime factorization.

Suppose now that we are given $a, b, c \in \mathbb{Z}$ and we want to solve $ax+by = c$, for $x, y \in \mathbb{Z}$. We first note that $(a, b) \mid a, b$, so for there to be any solutions, it is a necessary and sufficient condition that $(a, b) \mid c$:

- $c = (a, b)c' \implies ax'c' + by'c' = (a, b)c' = c$.
- $(a, b) \mid a, b \implies (a, b) \mid ax, by \implies (a, b) \mid ax + by = c$.

Now, let us write $c = d(a, b)$. We can also find integers m, n such that $(a, b) = ma + nb$, so we get that

$$c = d(a, b) = dma + dnb,$$

implying that $x = dm$, $y = dn$ is a solution. Now that we have a solution of the equation, suppose that we had a second solution: x', y'. Then, we find that

$$a(x - x') + b(y - y') = 0,$$

so we have $x' = x + r, y' = y + s$, where $ar + bs = 0$. Conversely, if $ar + bs = 0$, then $x+r, y+s$ is a solution to $ax+by = c$. Therefore, we are now reduced to finding solutions to

$$ar + bs = 0.$$

For any $r \in \mathbb{Z}$, to have an s such that $ar + bs = 0$, we need to have that

$$s = -\frac{ar}{b}.$$

Therefore, we need to determine for which integers r the expression above is an integers. This is precisely when $b \mid ar$. If we divide both sides by (a, b), we see that this happens if and only if $\frac{b}{(a,b)} \mid \frac{ar}{(a,b)}$. However, since $\left(\frac{b}{(a,b)}, \frac{a}{(a,b)}\right) = 1$, this happens if and only if $\frac{b}{(a,b)} \mid r$.

If we put this all together, we find that if x, y is one solution to $ax + by = c$, then we have infinitely many solutions of the form $x + n\frac{b}{(a,b)}, y - n\frac{a}{(a,b)}$, for all $n \in \mathbb{Z}$.

§1.2.i Congruences

Definition 1.2.4. Let $a, b, n \in \mathbb{Z}$. We say that a is **congruent** to b modulo n, $a \equiv b \pmod{n}$, if and only if there exists $n \mid a - b$.

Remark 1.2.5 — Note that $\equiv$ is an equivalent relation. The set of equivalence classes is denoted by $\mathbb{Z}/n$ (or $\mathbb{Z}/(n)$, $\mathbb{Z}/n\mathbb{Z}$).

Note that we have the following surjective map:

$$\begin{aligned} \mathbb{Z} &\to \mathbb{Z}/n \\ a &\mapsto a \pmod n =: [a]_n \end{aligned}$$

One can define addition and multiplication as follows:

$$\begin{aligned} [a]_n + [b]_n &= [a+b]_n \\ [a]_n[b]_n &= [ab]_n \end{aligned}$$

Theorem 1.2.6

There exists a unique commutative ring structure on $\mathbb{Z}/m$ which makes this map into a ring homomoprhism.

Proof. We just need to check the above map, with its addition and multiplication. □

Now, recall that for a ring R, we have the *units* of R, denoted $R^\times$, that consist of elements that have multiplicative inverses. Particularly focusing on when $R = \mathbb{Z}$, we will show that

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{[a]_n \mid (a,n) = 1\}.$$

Let a be an integer and suppose that $[a]_n$ is a unit. Then, there exists b such that $ab \equiv 1 \pmod n$; it is then apparent that $(a,n) = 1$. Conversely, if $(a,n) = 1$, there exists x, y such that $ax + ny = 1$. This means that $[a]_n[x]_n = [1]_n$, so $[a]_n$ is a unit. This shows the result above.

§1.2.ii Linear Congruence Equations

Now, suppose we fix $a, b, n \in \mathbb{Z}$, with $b \neq 0$, so that we want to solve

$$ax \equiv n \pmod b.$$

This is equivalent to finding solutions x, y to the equation

$$ax + by = n.$$

Particularly, we get solutions when $(a,b) \mid n$. Moreover, if we have that x is a solution, other solutions are of the form $x + r\frac{b}{(a,b)}$, where $r \in \mathbb{Z}$. Particularly, they form a single congruence class modulo $\frac{b}{(a,b)}$, or a total of (a,b) congruence classes modulo b.

§1.3 More Congruence Results

§1.3.i Chinese Remainder Theorem

Moving on, note that $\mathbb{Z}/n$ is an integral domain only when n is prime. Further, note that if we have a finite ring R that is an integral domain, it is a field, so $\mathbb{F}_p = \mathbb{Z}/p$ is the field of cardinality p. This particularly becomes necessary in proving Wilson's theorem.

Now, we consider the following situation in the setup for the *Chinese Remainder Theorem*. Let m, n be nonzero integers. If $a \equiv b \pmod{mn}$, then $a \equiv b \pmod n$ and

$a \equiv b \pmod m$. Therefore, a congruence class $\pmod{mn}$ determines a pair, consisting of a congruence class $\pmod m$ and a congruence class $\pmod n$. Formally, we have a map

$$\begin{aligned}\mathbb{Z}/mn &\to \mathbb{Z}/m \times \mathbb{Z}/n\\ [a]_{mn} &\mapsto ([a]_m, [a]_n).\end{aligned}$$

It can be checked that this map is a ring homomorphism. Now, it's natural to ask whether the converse is true: given a pair of congruence classes $\pmod m$ and $\pmod n$, is there a congruence class $\pmod{mn}$ giving rise to it? In general the answer will be no, but when $(m,n) = 1$, we have the Chinese Remainder Theorem:

Theorem 1.3.1 (Chinese Remainder Theorem)
Let m, n be integers with $(m,n) = 1$, and let a, b be integers. Then, there exists an integer x such that $x \equiv a \pmod m$ and $x \equiv b \pmod n$. Moreover, x is unique modulo mn.

Proof. We first prove uniqueness. If x, x' are two integers that have the required properties, then

$$x \equiv x' \pmod m \text{ and } x \equiv x' \pmod n,$$

implying that $x - x'$ is divisible by both m, n. Let us write $x - x' = cm$, for some integer c. We have that $n \mid cm$, and so $n \mid c$ because $(n,m) = 1$. If we write $c = dn$, we have the result:

$$x - x' = mnd \implies x \equiv x' \pmod{mn}.$$

For existence, first write $1 = nx + my$. Then, $anx + bmy$ is congruent to $a \pmod m$ and $b \pmod n$. □

Remark 1.3.2 — We can offer a non-constructive proof of existence as well. Note that in the uniqueness, we have shown that the map

$$\mathbb{Z}/mn\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$$

is injective. On the other hand, both the source and target of this map have same cardinality (of mn), we get surjectivity of this map, proving that the map is bijective, so it is a ring isomorphism.

As above, we can interpret the Chinese remain theorem as the map

$$\mathbb{Z}/mn\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$$

being an isomorphism of rings. In other words, to prove any statement about $\mathbb{Z}/mn\mathbb{Z}$ that can be formulated purely in ring-theoretic terms, it suffices to prove corresponding statements $\pmod m$ and $\pmod n$, provided that $(m,n) = 1$. What if $(m,n) \neq 1$? While the surjection still holds true, it is no longer injective.

Proposition 1.3.3
Let m, n be positive integers. Then, $\mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/\operatorname{lcm}(m,n) \times \mathbb{Z}/\gcd(m,n)$.

Proof. Let $d = \gcd(n, m)$. Then, Bézout tells us that there exists integers u, v such that $nx + my = d$. Now, we construct the map $\mathbb{Z}/\mathrm{lcm}(m,n) \times \mathbb{Z}/\gcd(m,n) \to \mathbb{Z}_m \times \mathbb{Z}_n$ by

$$(a + \mathrm{lcm}(n,m)\mathbb{Z}, b + \gcd(n,m)\mathbb{Z}) \mapsto (ua + \frac{m}{d}b + m\mathbb{Z}, va - \frac{n}{d}b + n\mathbb{Z}).$$

This is well defined group homomorphism. For the element on the left to be in the kernel, we require $ua + \frac{m}{d}b$ to be a multiple of m and $va - \frac{n}{d}b$ to be a multiple of n. Then, note that we have:

$$\frac{n}{d}\left(ua + \frac{m}{d}b\right) + \frac{m}{d}(va - \frac{n}{d}b) = \frac{nu + vm}{d}a = a$$

This is a multiple of $\frac{mn}{d} = \mathrm{lcm}(m,n)$. Thus, $a \equiv 0 \pmod{\mathrm{lcm}(m,n)}$. Then, $\frac{m}{d}b$ must be a multiple of m and $\frac{n}{d}b$ must be a multiple of n. This tells us that b must be a multiple of d, so $b \equiv 0 \pmod d$. Thus, our kernel is trivial, and so we have an injective map between groups of the same order, so this is an isomorphism. □

Definition 1.3.4. Two ideal $\mathfrak{a}, \mathfrak{b}$ of a ring R are **coprime** if $\mathfrak{a} + \mathfrak{b} = R$. This is equivalent to the existence of $x \in \mathfrak{a}, y \in \mathfrak{b}$ such that $x + y = 1$.

Now let R be a ring and let $\mathfrak{a}_1, \ldots, \mathfrak{a}_n$ be ideals of R. Consider the homomorphism

$$\phi \colon R \to \prod_{i=1}^{n} (R/\mathfrak{a}_i)$$

given by the rule $x \mapsto (x + \mathfrak{a}_1, \ldots, x + \mathfrak{a}_n)$. Then, we have the following:

Lemma 1.3.5

We have the following:

1. If $\mathfrak{a}_i$ and $\mathfrak{a}_n$ are coprime whenever $i \neq n$, then $\prod_{i=1}^{n} \mathfrak{a}_i$ and $\mathfrak{a}_n$ are coprime.
2. If $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$, then $\prod \mathfrak{a}_i = \cap \mathfrak{a}_i$.
3. The map ϕ is injective if and only if $\cap \mathfrak{a}_i = (0)$
4. The map ϕ is surjective if and only if $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$.

Proof. 1. Indeed, for every $i = 1, \ldots, n-1$, there are $x_i \in \mathfrak{a}_i$ and $y_i \in \mathfrak{a}_n$ such that $x_i + y_i = 1$, by assumption. Then,

$$1 = \prod_{i=1}^{n} (x_y + y + i) = x_1 \ldots x_{n-1} + \sum_{j=1}^{n-1} y_j \prod_{i \neq j} (x_i + y_j),$$

and clearly,

$$x_1 \ldots x_{n-1} \in \prod_{i=1}^{n-1} \mathfrak{a}_i \text{ and } \sum_{j=1}^{n-1} y_j \prod_{i \neq j} (x_i + y_j) \in \mathfrak{a}_n.$$

2. As $n = 1$ is obvious, assume that $n = 2$. As $\mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}_1, \mathfrak{a}_2$, we have $\mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}_1 \cap \mathfrak{a}_2$. Next, we show the reverse inclusion. Let $z \in \mathfrak{a}_1 \cap \mathfrak{a}_2$. By assumption, there are $x \in \mathfrak{a}_1$ and $y \in \mathfrak{a}_2$ such that $x + y = 1$. Then,

$$z = z(x + y) = zx + zy \in \mathfrak{a}_2\mathfrak{a}_1 + \mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}_1\mathfrak{a}_2.$$

We cover the case for $n \geq 3$ via induction on n. By the 1., the ideals $\prod_{i=1}^n \mathfrak{a}_i$ and $\mathfrak{a}_n$ are coprime, so

$$\prod_{i=1}^{n} \mathfrak{a}_i = \prod_{i=1}^{n} \mathfrak{a}_i \cdot \mathfrak{a}_n = \prod_{i=1}^{n} \mathfrak{a}_i \cap \mathfrak{a}_n$$

by the case $n = 2$, which we've already proven. Then, the inductive hypothesis tells us that

$$\prod_{i=1}^{n} \mathfrak{a}_i \cap \mathfrak{a}_n = \bigcap_{i=1}^{n} \mathfrak{a}_i \cap \mathfrak{a}_n = \bigcap_{i=1}^{n} \mathfrak{a}_i.$$

3. For every $i = 1, \dots, n$, let $\phi_i \colon R \to R/\mathfrak{a}_i$ be the homomorphism given by the rule $x \mapsto x + \mathfrak{a}_i$. Clearly,

$$\ker(\phi) = \bigcap_{i=1}^{n} \ker(\phi_i) = \bigcap_{i=1}^{n} \mathfrak{a}_i.$$

4. We do this for $n = 2$ first. If $\mathfrak{a}_1$ and $\mathfrak{a}_2$ are coprime, then there are $x \in \mathfrak{a}_1$ and $y \in \mathfrak{a}_2$ such that $x + y = 1$. Now, let $a, b \in R$ be arbitrary. Then,

$$bx + ay \equiv ax + by \equiv a(x + y) \equiv a \pmod{\mathfrak{a}_1},$$

and

$$bx + ay \equiv bx + by \equiv b(x + y) \equiv b \pmod{\mathfrak{a}_2},$$

so ϕ is surjective. On the other hand, if ϕ is surjective, there is an $x \in R$ such that $x \equiv 0 \pmod{\mathfrak{a}_1}$ and $x \equiv 1 \pmod{\mathfrak{a}_2}$. Then, $x \in \mathfrak{a}_1$ and $y = 1 - x \in \mathfrak{a}_2$, and since $x + y = 1$, we get that $1 \in \mathfrak{a}_1 + \mathfrak{a}_2$, and hence $\mathfrak{a}_1 + \mathfrak{a}_2 = R$. We now prove the case $n \geq 3$ by induction on n. If ϕ is surjective then the map

$$R \to R/\mathfrak{a}_i \times R/\mathfrak{a}_j$$

is also surjective for every $i \neq j$, so $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime by the above. Now, assume that $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$, and let $b_1 \dots, b_n \in R$ be arbitrary. By the inductive hypothesis, there is an $x \in R$ such that $x \equiv b_i \pmod{\mathfrak{a}_i}$ for $i = 1, \dots, n-1$. By 1., the ideal $\prod\limits_{i=1}^{n} \mathfrak{a}_i$ and $\mathfrak{a}_n$ are coprime, so there is a $b \in R$ such that

$$b \equiv x \pmod{\prod_{i=1}^{n-1} \mathfrak{a}_i}$$

and $b \equiv b_n \pmod{\mathfrak{a}_n}$ by the case $n = 2$, which we've already proved. since $\prod\limits_{i=1}^{n-1} \mathfrak{a}_i \subseteq a_j$, for every $j = 1, \dots, n-1$, we get that

$$b \equiv x \equiv b_j \pmod{\mathfrak{a}_j},$$

for every $j = 1, \dots, n-1$ as well. □

Remark 1.3.6 — Let R be now an integral domain. We say that two elements $0 \neq a, b \in R$ are **coprime** if their g.c.d. exists and it is associated to $1 \in R$. When R is an Euclidean domain this is equivalent to the principal ideals (x) and (y) being coprime. As a corollary we get the usual Chinese remainder theorem for any number of moduli over the integers.

§1.3.ii Wilson's Theorem

Theorem 1.3.7 (Wilson's Theorem)

Let p be prime. Then,
$$(p-1)! \equiv -1 \pmod p$$

Proof. Consider the equivalence class of $(p-1)! \pmod p$. Since inverses exist in p, there is a lot of cancellation that occurs. Particularly, for any a such that $p \nmid a$, there exists a unique integer a^{-1} such that $aa^{-1} \equiv 1 \pmod p$. Particularly, $a = a^{-1}$ or a and a^{-1} are distinct modulo p. In the second case, the pair of congruence classes $\{a, a^{-1}\}$ disappears from the product. Therefore, this product is equal to the product of congruence classes a such that $a = a^{-1}$. These are the classes such that $a^2 \equiv 1 \pmod p$. Note that this only happens when
$$(a+1)(a-1) \equiv 0 \pmod p,$$
and since $\mathbb{Z}/p\mathbb{Z}$ is a field, this is when one of the factors on the LHS is zero. Thus, the classes that are their own inverses are $\pm 1 \pmod p$, so we're done, as $(p-1)!$ is the product of these classes. $\square$

2 Arithmetic Functions

Definition 2.0.1. A function $\phi \colon \mathbb{N} \to \mathbb{C}$ is an **arithmetic function**.

§2.1 Euler's Function and Euler's Theorem

Definition 2.1.1. We denote by $\phi(n)$ to be the number of elements in the set $\{a \mid 0 < a < n, (a,n) = 1\}$. In other words, $\phi(n)$ counts the number of integers between 0 and $n-1$ that are coprime to n.

> **Remark 2.1.2** — Note that $\phi(n)$ denotes the order order of $(\mathbb{Z}/n)^\times$.

If we note that $\mathbb{Z}/1 = 0$, we get that $\phi(1) = 1$.

> **Theorem 2.1.3** (Euler's Theorem)
> Let a be an integer with $(a,n) = 1$. Then, $a^{\phi(n)} \equiv 1 \pmod{n}$.

Proof. Let a be an integer such that $(a,n) = 1$ and L be the product

$$L = \prod_{\substack{1 \le q < n \\ (q,n)=1}} (aq).$$

On one hand, we have

$$L = a^{\phi(n)} \prod_{\substack{1 \le q < n \\ (q,n)=1}} q.$$

On the other, we know that for each q' with $1 \le q' < n$ such that $(q',n) = 1$, the equation $aq \equiv q' \pmod{n}$ has a unique solution modulo n, since $(a,n) = 1$. Therefore, there is a unique q, with $1 \le q < n$ such that $aq \equiv q' \pmod{n}$. Therefore, we can write

$$L = \prod_{\substack{1 \le q < n \\ (q,n)=1}} (aq) \equiv \prod_{\substack{1 \le q' < n \\ (q',n)=1}} q' \pmod{n}.$$

Thus, equating both equations expressing L, we have

$$a^{\phi(n)} \prod_{\substack{1 \le q < n \\ (q,n)=1}} q \equiv \prod_{\substack{1 \le q' < n \\ (q',n)=1}} q' \pmod{n},$$

and cancellation of the product provides the desired result. Note that we can cancel, since we can find inverses for each element. $\square$

> **Remark 2.1.4** — We can also apply Lagrange's theorem to $(\mathbb{Z}/n)^\times$ to prove Euler's theorem.

Corollary 2.1.5 (Fermat's Little Theorem)

Let p be prime and $p \nmid a$. Then, $a^{p-1} \equiv 1 \pmod p$.

Proof. Follows for when $n = p$ for Euler's theorem. □

Definition 2.1.6. A function f on the positive integers is **multiplicative** if for all positive integers m, n such that $(m, n) = 1$, we have $f(mn) = f(m)f(n)$. We say f is **strictly multiplicative** if for any pair of positive integers m, n we have $f(mn) = f(m)f(n)$.

Remark 2.1.7 — Note that from its definition, a multiplicative function is determined by its values on prime powers.

Theorem 2.1.8

The Euler function ϕ is multiplicative.

Proof. This follows by the Chinese remainder theorem. If m and n are positive integers with $(m, n) = 1$, we have an isomorphism of rings: $\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$, and hence an isomorphism of groups:

$$(\mathbb{Z}/mn)^\times \cong (\mathbb{Z}/m)^\times \times (\mathbb{Z}/n)^\times.$$

This result then follows. □

Remark 2.1.9 — Note that ϕ isn't strictly multiplicative: $\phi(2) = 1$ and $\phi(4) = 2$.

Theorem 2.1.10

We can represent Euler's function as follows:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

Proof. We start with prime powers, so for prime p, we have $(a, p^r) = 1$ if and only if $p \nmid a$. Therefore, it follows that $\phi(p^r)$ denotes the number of integers between 0 and $p^r - 1$ that aren't divisible by p. There are p^{r-1} number divisible by p, so we have that

$$\phi(p^r) = p^r - p^{r-1} = p^r \left(1 - \frac{1}{p}\right).$$

Then, using the multiplicativity of ϕ, we get that

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

□

Theorem 2.1.11

Let n be a positive integer. Then,

$$\sum_{d|n} \phi(d) = n.$$

Proof. Define two functions f, g as follows:

$$f(n) = \sum_{d|n} \phi(d)$$
$$g(n) = n.$$

Clearly g is strictly multiplicative, and so is multiplicative. Thus, if we show that f is multiplicative, and $f(p^k) = g(p^k)$ for all primes p and integers k, we must have $f = g$ and the theorem is proved. First, we show $f(p^k) = g(p^k)$.

$$f(p^k) = \sum_{d|p^k} \phi(d) = \sum_{i=0}^{k-1} \phi(p^i) = 1 + (p-1) + \cdots + (p^k - p^{k-1}) = p^k.$$

It remains to prove multiplicativity of f. Suppose that m and n are relatively prime integers, so that we have

$$f(mn) = \sum_{d|mn} \phi(d).$$

Note that for any divisor d of mn, if we set $t = (d, m)$ and $u = (d, n)$, then we have $t \mid m, u \mid n$, and $tu = d$. Moreover, this gives a bijection between divisors d of mn and pairs (t, u) consisting of a divisor t of m and a divisor u of n. We can thus rewrite the above equation as

$$f(mn) = \sum_{t|m} \sum_{u|n} \phi(tu).$$

Moreover, for any t, u appearing in the sum, t and u are relatively prime (since any common factor would divide both m and n). We thus have:

$$f(mn) = \sum_{t|m} \sum_{u|n} \phi(t)\phi(u) = \left(\sum_{t|m} \phi(t)\right) \left(\sum_{u|n} \phi(t)\right)$$

□

We offer an alternative proof to the previous theorem below, which is more conceptual.

Alternative Proof. Note that $(m, n) = 1$ if and only if the additive subgroup of $\mathbb{Z}/n$ generated by m is equal to all of $\mathbb{Z}/n$. In other words, $\phi(n)$ counts the elements of the cyclic group $\mathbb{Z}/n$ that have order exactly n. Another way of saying this is that $\phi(n)$ is the number of elements of order n in a cyclic group of order n.

Now suppose C is a cyclic group of order divisible by n (but not necessarily equal to n). Then C contains a cyclic subgroup C' of order n. Moreover, this subgroup C' contains every element of C of order dividing n. In particular, the number of elements of C of order n is equal to the number of elements of C' of order n, and this number is of course $\phi(n)$. Thus $\phi(n)$ counts the number of elements of order n in any cyclic group of order divisible by n.

If we then consider the sum

$$\sum_{d|n} \phi(d).$$

The above discussion shows that $\phi(d)$ is the number of elements of order d in a cyclic group C of order n. Since every element of C order d for some $d \mid n$, the total number of such elements is just the number of elements C, so the sum must equal n. □

§2.1.i Primitive Roots

Proposition 2.1.12

Let a be an integer with $(a, n) = 1$, and let k be the order of a mod n. If $a^d \equiv 1 \pmod{n}$ then k divides d.

Proof. Write $d = qk + r$ with q, r integers and $0 \leq r < k$. Then,

$$1 \equiv a^d = (a^k)^q a^r \pmod{n}.$$

Since $a^k \equiv 1 \pmod{n}$, it follows that $a^r \equiv 1 \pmod{n}$. Since $r < k$, r cannot be positive (by the definition of order), so $r = 0$ and $d = qk$. □

It now follows from Euler's theorem that for any a with $(a, n) = 1$, the order of a mod n divides $\phi(n)$.

Definition 2.1.13. An integer a with $(a, n) = 1$ is a **primitive root** mod n if the order of a mod n is exactly $\phi(n)$.

Example 2.1.14

Primitive roots need not exist every modulo n. For instance, if $n = 8$, then $3, 5, 7$ have order 2 mod 8, and 1 has order 1. Since $\phi(8) = 4$, there are no primitive roots mod 8.

Lemma 2.1.15

Let R be a ring, $P(X)$ be a polynomial in X with coefficients in R. If $P(\alpha) = 0$, then there exists a polynomial $Q(X)$ with coefficients in R such that $P(X) = Q(X)(X-\alpha)$.

Proof. We proceed by induction on the degree of P, the degree zero case being clear. Suppose the result is true for polynomials of degree $d-1$, and let $P(X)$ have degree d. If the leading term of $P(X)$ is cX^d, let $S(X) = P(X) - cX^{d-1}(X - \alpha)$. We have $S(\alpha) = 0$, and $S(X)$ has degree $d - 1$, so there exists $T(X)$ with coefficients in R such that $S(X) = T(X)(X - \alpha)$. Then, $P(X) = [cX^d + T(X)](X - \alpha)$. □

Theorem 2.1.16

Let F be a field, and $P(X)$ be a polynomial of degree d with coefficient in F. Then, $P(X)$ has at most d distinct roots in F.

Proof. We again proceed by induction on d; the case $d = 0$ is clear. If $P(X)$ has degree d and α is a root, we can write $P(X) = (x - \alpha)Q(X)$. Now if $P(\beta) = 0$, then $(\beta - \alpha)Q(\beta) = 0$, so (since D is a field) either $\beta = \alpha$ or β is a root of $Q(X)$. By the inductive hypothesis $Q(X)$ has at most $d - 1$ roots in F, so P has at most d roots as claimed. □

Corollary 2.1.17

Let p be a prime, and let $d \mid p - 1$. Then the polynomial $x^d - 1$ has exactly d roots mod p.

Proof. By Fermat's little theorem, $[1]_p, \dots, [p-1]_p$ are all roots of $x^{p-1} - 1$ mod p. Thus, $x^{p-1} - 1$ has exactly $p - 1$ roots. Now fix d that divides $p - 1$ and write $x^{p-1} - 1 = (x^d - 1)Q(X)$ for a polynomial $Q(X)$. $Q(X)$ has integer coefficients, so we can view it as a polynomial mod p. Now, $x^{p-1} - 1$ has $p - 1$ roots, x^d has at most d roots, and $Q(X)$ has at most $p - q - d$ roots, so we must have equality. That is, $x^d - 1$ has d roots mod p. □

Remark 2.1.18 — Another way of stating the corollary is to say that for any $d \mid p - 1$, there are exactly d elements of $(\mathbb{Z}/p)^\times$ whose order divides d.

Theorem 2.1.19 (Existence of Primitive Roots)

Let p be a prime. then, for any $d \mid p - 1$, there are exactly $\phi(d)$ elements of order d in $(\mathbb{Z}/p)^\times$. Particularly, there are $\phi(p - 1)$ primitive roots mod p.

Proof. We prove this by strong induction on d; the case $d = 1$ is clear. Fix d. The inductive hypothesis tells us that for any $d' \mid d$ with $d' < d$, there are $\phi(d')$ elements of exact order d'. On the other hand, there are a total of d elements of $(\mathbb{Z}/p)^\times$ of order dividing d. Thus, the number of elements of $(\mathbb{Z}/p)^\times$ of order d is

$$d - \sum_{\substack{d' \mid d \\ d' < d}} \phi(d'),$$

which is precisely $\phi(d)$. □

Finite Subgroups of Fields

We can now prove the following theorem, which is a specific case of the one following it.

Theorem 2.1.20

$(\mathbb{F}_p)^\times \cong \mathbb{Z}/p - 1$; that is, it's a cyclic group.

Theorem 2.1.21

Let F be a field. If $G \subseteq F^\times$ is a finite subgroup, then G is cyclic.

Proof. Let $\mu_d = \{x \in F^\times \mid x^d = 1\}$, which are the dth roots of unity. Note that $\mu_d \subset F^\times$, as $|\mu_d| \leq d$. $G \cap \mu_d \subseteq \mu_d$. Then we use induction on the order of the group. By the inductive hypothesis, $G \cap \mu_d$ is a cyclic group. We now count the number of elements that have order d in $G \cap \mu_d$. If $\alpha \in G$ such that $\mathrm{ord}(\alpha) = d$, then $\alpha^d = 1$, so $\alpha \in \mu_d$. We may count the number of elements of $G \cap \mu_d$ that are cyclic.

- If $|G \cap \mu_d| < d$, then the number of such elements is empty.
- If $|G \cap \mu_d| = d$, then the number of such elements is $\phi(d)$.

We now count the number of elements of order n in G.

$$|G| - \sum_{\substack{d|n \\ d<n}} \underbrace{|\{\alpha \in G \mid \mathrm{ord}(\alpha) = d\}|}_{=\phi(d)} \geq n - \sum_{\substack{d|n \\ d<n}} \phi(d) = n - (n - \phi(n)) = \phi(n) > 0.$$

Therefore, there is at least one element with order n; that is $G \cap \mu_n \neq \emptyset$. Thus, G is cyclic of order n. $\square$

Theorem 2.1.22

$(\mathbb{Z}/n)^\times$ is cyclic if and only if $n = 2, 4, p^\alpha, 2p^\alpha$, where p is odd.

Proof. This follows by two facts from group theory. First, that a product $G = C_{n_1} \times \ldots C_{n_k}$ is not cyclic unless $n_1, \ldots, n_k$ are coprime. Secondly, we require that every subgroup of a cyclic subgroup is cyclic. This then follows by the fact that $(\mathbb{Z}/2^n)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{n-2}$. $\square$

Proposition 2.1.23

$(\mathbb{Z}/2^n)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{n-2}$.

Proof. We claim that $\mathrm{ord}_{2^n}(5) = 2^{n-2}$. To prove this, we will show that $5^{2^{n-3}} \not\equiv 11 \pmod{2^k}$. For this, we use the binomial theorem:

$$5^{2^{n-3}} = (2^2 + 1)^{2^{n-3}} = 1 + 2^{n-1} + \sum_{i \geq 2} \binom{2^{n-3}}{i} 2^{2i}$$

We claim this is congruent to $1 + 2^{n-1} \pmod{2^n}$ (and so not congruent to 1). However, this is clear, as all the terms remaining in the sum are divisible by 2^n. Now the subgroup generated by 5 in $(\mathbb{Z}/2^n)^\times$ is of order 2^{n-2}. Note that we've shown above that

$$5^{2^{n-3}} \equiv 1 + 2^{n-1} \pmod{2^n},$$

which is not equivalent to -1. Thus, -1 doesn't lie in $\langle 5 \rangle$. Therefore, we get the desired group structure, as $-1 \cdot 5^j$, for $j = 0, 1, \ldots, 2^{n-2} - 1$ are different to $\langle 5 \rangle$. $\square$

Remark 2.1.24 — The above is a nice way to say that there exists a primitive root modulo n.

Exercise 2.1.25. Using primitive roots, show that $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic.

Solution: Let g be a primitive root modulo p. From here, we consider g modulo p^2. If $g^{p-1} \not\equiv 1 \pmod{p^2}$, then we're done. the order of g modulo p^2 must divide $p(p-1)$, but this order is not $p-1$; thus, it must be $p(p-1)$, so g is a primitive root modulo p^2; thus, $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic.

Now suppose that $g^{p-1} \equiv 1 \pmod{p^2}$. The claim is that $(g+p)^{p-1} \not\equiv 1 \pmod{p^2}$. Indeed, by freshman's dream, we have

$$(g+p)^{p-1} \equiv g^{p-1} + p(p-1)g^{p-2} \pmod{p^2}.$$

Since $(p-1)g^{p-2}$ is not divisible by p, we're done.

The claim now is that if the order of g modulo p^2 is $p(p-1)$, then for any higher powers of p, the order of g modulo p^m is $\phi(p^m) = p^{m-1}(p-1)$; this would imply that $(\mathbb{Z}/p\mathbb{Z})^m$ is cyclic. From here, we write $\mathrm{ord}_a(g)$ for the order of g modulo a. Suppose that $m \geq 2$. Note that $\mathrm{ord}_{p^2}(g) \mid \mathrm{ord}_{p^m}(g) \mid \phi(p^m)$. Therefore, $\mathrm{ord}_{p^m}(g) = p^s(p-1)$ for some $1 \leq s \leq m-1$. Therefore, the statement $\mathrm{ord}_{p^m}(g) = p^{m-1}(p-1)$ is equivalent to showing that $g^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m}$. Suppose that this holds for some m though. Then, considering modulo p^{m-1}, we have that $g^{p^{m-2}(p-1)} \equiv 1 \pmod{p^{m-1}}$. Therefore, we can write

$$g^{p^{m-2}(p-1)} = 1 + p^{m-1}t,$$

where $p \nmid t$. Then, we may apply the freshman's dream so we get that

$$g^{p^{m-1}(p-1)} \equiv (1+p^{m-1}t)^p \equiv 1 + p^m t \pmod{p^{m+1}},$$

so the statement holds for $m+1$, so we're done by induction. $\square$

§2.1.ii Discrete Logarithms

Recall that if g is a primitive root mod n, then by definition, the order of g is $\phi(n)$.It follows that the powers $g, g^2, \ldots, g^{\phi(n)}$ are all distinct mod n. This is because if $g^i \equiv g^j \pmod{n}$, then $g^{i-j} \equiv 1 \pmod{n}$. There are only $\phi(n)$ classes in $(\mathbb{Z}/n)^\times$, so this says that if g is a primitive root then very class in $(\mathbb{Z}/n)^\times$ is a power of g. In other words, the group $(\mathbb{Z}/n)^\times$ is a cyclic group of order $\phi(n)$, generated by g. For this reason, primitive roots are often called **generators**.

Another way to say this is that when a primitive root g exists mod n, the map

$$\begin{aligned} \mathbb{Z}/\phi(n) &\to (\mathbb{Z}/n)^\times \\ [a]_{\phi(n)} &\mapsto [g^a]_n \end{aligned}$$

is an isomorphism, from the additive group of $\mathbb{Z}/\phi(n)$ to$(\mathbb{Z}/n)^\times$. It thus has an inverse, which we call the *discrete logarithm* to the base g.

Definition 2.1.26. If g is a primitive root mod n, the **discrete logarithm** to the base g, denoted $\log_g$ is defined by $\log_g a = [k]_{\phi(n)}$, where k is an integer such that $g^k \equiv a \pmod{n}$.

One use of the discrete logarithm is to solve exponential equation mod n, if n admits a primitive root g. For instance, by applying $\log_g$ to both side of the equation $x^d \equiv a \pmod{n}$, we attain a linear congruence equation

$$d \log_g x \equiv \log_g a \pmod{\phi(n)},$$

and we can solve those with techniques which can be solved with familiar methods.

§2.2 Multiplicative Functions

Recall the definition of multiplicative functions.

§2.2.i Convolution

Definition 2.2.1. Given function $f, g\colon \mathbb{Z}_{>0} \to \mathbb{C}$, the **convolution** $f * g$ is defined as follows:

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right).$$

Proposition 2.2.2

The convolution has the following properties:

1. If f and g are multiplicative, then so is $f * g$.
2. Convolution is commutative: $f * g = g * f$, for all f, g.
3. Convolution is associative: $(f * g) * h = f * (g * h)$, for all f, g, h.
4. Let δ be the function defined by

$$\delta(n) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{for } n > 1. \end{cases}$$

Then, $f * \delta = f$.

Proof. 1. Let m and n be relatively prime. We then have

$$\begin{aligned}(f * g)(mn) = \sum_{d|mn} f(d) g\left(\frac{mn}{d}\right) &= \sum_{t|m} \sum_{u|n} f(tu) g\left(\frac{mn}{tu}\right) \\ &= \left(\sum_{t|m} f(t) g\left(\frac{m}{t}\right)\right)\left(\sum_{u|n} f(u) g\left(\frac{n}{u}\right)\right) = (f * g)(m)(f * g)(n).\end{aligned}$$

Note that in the above, we set $t = (d, m)$ and $u = (d, n)$ so that we get a bijection between divisors d of mn and pairs (t, u) consisting of a divisor t of m and u of n.

2. We have $(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right)$. If we let $e = \frac{n}{d}$, the latter is equal to

$$\sum_{e|n} f\left(\frac{n}{e}\right) g(e) = (g * f)(n).$$

3. We have the two equations:

$$((f*g)*h)(n) = \sum_{d|n}(f*g)(d)h\left(\frac{n}{d}\right) = \sum_{d|n}\sum_{e|d} f(e)g\left(\frac{d}{e}\right)h\left(\frac{n}{d}\right)$$
$$(f*(g*h))(n) = \sum_{d'|n} f(d')(g*h)\left(\frac{n}{d'}\right) = \sum_{d'|n}\sum_{e'|\frac{n}{d'}} f(d')g(e')h\left(\frac{n}{d'e'}\right).$$

These two are related by the change of variables $e = d'$, $d = d'e'$, which has inverse $d' = e$, $e' = \frac{d}{e}$.

4. We have
$$(f*\delta)(n) = \sum_{d|n} f(d)\delta\left(\frac{n}{d}\right) = f(n),$$

since the only term of the sum for which $\delta\left(\frac{n}{d}\right)$ is nonzero is $d = n$. □

Remark 2.2.3 — This convolution *almost* makes the set of multiplicative functions into an abelian group. However, the set of multiplicative functions form a commutative monoid with identity. Particularly, the operation is commutative, associative, and has identity, Typically, however, a function f will not have an inverse under the convolution operation.

Remark 2.2.4 — Note that we can write the function δ as
$$\delta(n) = \left\lfloor \frac{1}{n} \right\rfloor.$$

Example 2.2.5

The function $\mathbf{1}$ that takes the value 1 at all integers n is multiplicative.

Example 2.2.6

The function f_k defined by $f_k(n) = n^k$ is multiplicative.

Example 2.2.7

The function $\sigma_k = f_k * \mathbf{1}$, which is the sum
$$\sigma_k(n) = \sum_{d|n} d^k,$$

is multiplicative, since it is built by convolution of two multiplicative functions.

Example 2.2.8

The **Möbius function** μ is multiplicative. It is defined as follows:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } a_1 = a_2 = \cdots = a_k = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$.

Note that the significance of the Möbius function comes from how it acts as an inverse the function $\mathbf{1}$ under convolution.

Theorem 2.2.9 (Möbius Inversion)

The convolution $\mu * \mathbf{1}$ is δ.

Proof. Both sides are multiplicative, so it suffices to check this for prime powers.

$$(\mu * \mathbf{1})(p^k) = \sum_{d \mid p^k} \mu(d) = \sum_{i=0}^{k} \mu(p^i) = 1 + (-1) + 0 + \cdots + 0 = 0 = \delta(p^k).$$

□

Corollary 2.2.10

The convolution $\sigma_k * \mu$ is f_k.

Proof. We have as follows:

$$\sigma_k * \mu = (f_k * \mathbf{1}) * \mu = f_k * (\mathbf{1} * \mu) = f_k * \delta = f_k.$$

□

Theorem 2.2.11

We have $\phi = f_1 * \mu$.

Proof. It suffices to check for prime powers, as both sides are multiplicative. We seek for $\phi(p^k) = p^k - p^{k-1}$.

$$(f_1 * \mu)(p^k) = \sum_{i=0}^{k} p^k \mu(p^{k-i}) = 0 + \cdots + 0 + (-p^{k-1}) + p^k = p^k - p^{k-1}.$$

□

We attain another proof of the identity:

Corollary 2.2.12

Let $n > 1$. Then,
$$\sum_{d|n} \phi(d) = n.$$

Proof. We have $\phi = f_1 * \mu$. Convolving both sides with $\mathbf{1}$, we attain
$$\phi * \mathbf{1} = (f_1 * \mu) * \mathbf{1} = f_1 * \delta = f_1.$$
However, $f_1(n) = n$, and $(\phi * \mathbf{1})(n) = \sum_{d|n} \phi(d)$. □

§2.2.ii Perfect Numbers

Definition 2.2.13. A positive integer n is **perfect** if the sum of its divisors, other than n itself, is equal to n.

Example 2.2.14

6 is a perfect number, since $1 + 2 + 3 = 6$.

Remark 2.2.15 — In terms of multiplicative functions, n is perfect if and only if $\sigma_1(n) = 2n$.

Abuse of Notation 2.2.16. For brevity, we denote by $\sigma := \sigma_1$ for this subsection.

We observe that if n is an integer such that $2^n - 1$ is prime, then $2^{n-1}(2^n - 1)$ is perfect. Indeed, we have
$$\sigma(2^{n-1}(2^n - 1)) = \sigma(2^{n-1})\sigma(2^n - 1),$$
and
$$\sigma(2^{n-1}) = 1 + 2 + \cdots + 2^{n-1} = 2^n - 1$$
$$\sigma(2^n - 1) = 1 + (2^n - 1) = 2^n.$$
Thus, $\sigma(2^{n-1}(2^n - 1)) = 2^n(2^n - 1)$. Every perfect number is of this form.

Definition 2.2.17. A prime of the form $2^n - 1$ is called a **Mersenne prime**.

Note that a Mersenne prime occurs only for prime n. This is because if $n = ab$, then
$$2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 1).$$

Definition 2.2.18. A prime of the form $2^{2^n} + 1$ is called a **Fermat prime**.

Exercise 2.2.19. If $2^{2^n} + 1$ is prime, then n is a 2-power.

Theorem 2.2.20

Let n be an even perfect number, and write $n = 2^k m$ with $k \geq 1$ and m odd. Then, $m = 2^{k+1} - 1$ and is prime. Particularly, every even perfect number arises from a Mersenne prime from the recipe above.

Proof. If n is perfect, then $\sigma(n) = 2n$. In terms of m and k, we have on one hand $\sigma(2^k m) = 2^{k+1}m$. On the other, we have $\sigma(2^k m) = \sigma(2^k)\sigma(m) = (2^{k+1} - 1)\sigma(m)$. Thus, we have $2^{k+1}m = (2^{k+1} - 1)\sigma(m)$. Rearrangement provides

$$\sigma(m) = m\frac{2^{k+1}}{2^{k+1} - 1} = m + \frac{m}{2^{k+1} - 1}.$$

Since $\sigma(m)$ and m are integers, so is $\frac{m}{2^{k+1}-1}$. Moreover, then $\frac{m}{2^{k+1}-1}$ divides m. Thus, the right hand side expresses $\sigma(m)$ as a sum of two divisors of m. Since $\sigma(m)$ is the sum of all the divisors of m, it must be that m and $\frac{m}{2^{k+1}-1}$ are the only two divisors of m. Therefore, $\frac{m}{2^{k+1}-1} = 1$, so $m = 2^{k+1} - 1$. Moreover, since m has exactly two divisors, m is prime. This implies that $k + 1$ is prime; if $k + 1 = ab$, then $(2^a - 1)$ would divide $2^{ab} - 1$, so m wouldn't be prime. Therefore, we have that $k + 1 = p$, and so $n = 2^{p-1}(2^p - 1)$. $\square$

Remark 2.2.21 — By contrast, it is still not known whether odd perfect numbers exist although it has been shown that such a number has many hard to satisfy properties, and it is strongly suspected that no odd perfect numbers exist.

§2.3 Quadratic Residues

Definition 2.3.1. Let p be an odd prime and a an integer not divisible by p. We say that a is a **quadratic residue** mod p if there exists an integer d with

$$d^2 \equiv a \pmod{p}.$$

If no such d exists, we say that a is a **quadratic non-residue** mod n.

Remark 2.3.2 — By convention, integers a divisible by p are neither quadratic residues nor quadratic non-residues mod p.

Proposition 2.3.3

Let a, b be integers and p be an odd prime with $(a, p) = (b, p) = 1$. Then:

- If a and b are quadratic residues mod p, then so is ab.
- If a is a quadratic residue mod p and b is a quadratic non-residue mod p, then ab is a quadratic non-residue mod p.
- If a and b are quadratic non-residues mod p, then ab is a quadratic residue mod p.

Proof. For the first claim, if $a \equiv d^2 \pmod{p}$ and $b = c^2 \pmod{p}$, then $ab \equiv (cd)^2 \pmod{p}$. For the second, if $a \equiv d^2 \pmod{p}$ and $ab \equiv c^2 \pmod{p}$, then we have $[b]_p = [a]_p^{-1}[ab]_p = ([d]_p^{-1}[c]_p)^2$. The third requires more than just the group structure on $(\mathbb{Z}/p)^\times$. We select a primitive root g mod p and write $a \equiv g^r \pmod{p}, b \equiv g^s \pmod{p}$. Then, r and s are odd since a and b are non-residues. But then $ab \equiv g^{r+s} \pmod{p}$, and $r+s$ is even. □

Definition 2.3.4. The **Jacobi symbol** $\left(\frac{a}{p}\right)$, for prime p and a an integer, is defined as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue mod } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue mod } p, \\ 0 & \text{if } p \mid a. \end{cases}$$

Theorem 2.3.5

The Jacobi symbol is completely multiplicative.

The proposition above then tells us the following:

Theorem 2.3.6

The map

$$(\mathbb{Z}/p)^\times \to \{\pm 1\}$$

given by

$$[a]_p \mapsto \left(\frac{a}{p}\right)$$

is a surjective group homomorphism if p is an odd prime.

The existence of primitive roots gives us an easy description of this map:

Theorem 2.3.7 (Euler's Criterion)

Let p be an odd prime, and a an integer such that $p \nmid a$. Then,

$$a^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

Proof. If $p \mid a$, this is clear. Otherwise, note that $\left(a^{\frac{p-1}{2}}\right)^2 \equiv 1 \pmod{p}$ by Fermat's little theorem. Therefore,

$$a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}.$$

If a is a quadratic residue mod p, then $a \equiv d^2 \pmod{p}$ for some d, and then $a^{\frac{p-1}{2}} \equiv d^{p-1} \equiv 1 \pmod{p}$. On the other hand, if a is a quadratic non-residue, then $a \equiv g^r \pmod{p}$, for some odd integer r and primitive root g mod p. Then, $a^{\frac{p-1}{2}} \equiv g^{\frac{r(p-1)}{2}} \pmod{p}$. Since r is odd, $\frac{r(p-1)}{2}$ isn't divisible by $p-1$. Thus, because g has order $p-1$, $g^{\frac{r(p-1)}{2}}$ is not congruent to 1 mod p, so it must be -1 mod p. □

Proposition 2.3.8

The integer -1 is a quadratic residue mod p if and only if $p \equiv 1 \pmod 4$, and a quadratic non-residue if $p \equiv 3 \pmod 4$. That is,

$$\left(\frac{-1}{p}\right) = 1, \iff p \equiv 1 \pmod 4.$$

Proof. This follows by Euler's criterion for $a = -1$.

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}.$$

$\square$

Remark 2.3.9 — We can also reduce this to a group-theoretical problem of whether there exists an element of order 4 in $\mathbb{F}_p^\times$. That is, $\left(\frac{-1}{p}\right) = 1 \iff -1$ is a square mod $p \iff \exists$ elements of order 4 in $\mathbb{F}_p^\times$.

Proposition 2.3.10

If $G \cong \mathbb{Z}/n$, then G has an element of order d if and only if $d \mid n$.

Proof. ($\implies$): For any group G, if $x \in G$ with $\mathrm{ord}(x) = d$, then $d \mid |G|$.

($\impliedby$): G is cyclic, we actually counted elements of order d; there are $\phi(d)$ of them. $\square$

Proposition 2.3.11 (Gauss's Lemma)

Let p be an odd prime.

- $\left(\frac{2}{p}\right) = 1$, if $p \equiv 1, 7 \pmod 8$.
- $\left(\frac{2}{p}\right) = -1$, if $p \equiv 3, 5 \pmod 8$

Proof. Let $q = \frac{p-1}{2}$, and set $Q = (2)(4)(6)\dots(p-1) = 2^q \cdot q!$. Reduce all the factors in the product defining Q mod p so that they lie in between $-q$ and q, i.e., subtract p from every factor greater than q. Let Q' be the resulting product $(2)(4)(8)\dots(-3)(-1)$. We have $Q' \equiv Q \pmod p$. On the other hand, the factors in the product defining Q' are the even integers from 1 to q and the negatives of the odd integers from 1 to q. We thus have $2^q \cdot q! \equiv (-1)^r(q!) \pmod p$, so $2^q \equiv (-1)^r \pmod p$. The result then follows by noting that r is even precisely when $p \equiv 1, 7 \pmod 8$, and invoking Euler's criterion. $\square$

Remark 2.3.12 — Note that since the Jacobi symbol is completely multiplicative, $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$, to answer this question in full generality it suffices to answer it for $a = -1$, $a = 2$, and for a an odd primes, which will be answered by the law of quadratic reciprocity (2.3.17).

§2.3.i The Law of Quadratic Reciprocity

The main goal here is to prove the law of quadratic reciprocity (2.3.17). We try to compute $\left(\frac{p}{q}\right)$ in the same way as $\left(\frac{2}{p}\right)$ above; that is, we consider the product $(p)(2p)(3p)\dots(\frac{q-1}{2}p)$. On one hand, this is equal to $p^Q Q!$, where $Q = \frac{q-1}{2}$. On the other, we can reduce this product mod q. Particularly, for $1 \le i \le Q$, let s_i be the unique integer between $-Q$ and Q such that $s_i \equiv pi \pmod{q}$. Then,

$$p^Q Q! = (p)(2p)(3p)\dots(Qp) \equiv \prod_{i=1}^{Q} s_i \pmod{q}.$$

On the other hand, we have the following lemma:

Lemma 2.3.13

For each $1 \le j \le Q$, there exists a unique i between 1 and Q such that $s_i = \pm j$.

Proof. It suffices to show that s_i is never 0, and that s_i is never equal to $\pm s_j$, for $i \neq j$. For the first of these, note that if $s_j = 0$, then $pi \equiv 0 \pmod{q}$. This is not possible, since p and i are both invertible mod q. Suppose we have $1 \le i < j \le Q$, with $s_i = \pm s_j$. Then, either $p(i+j) \equiv 0 \pmod{q}$ or $p(j-i) \equiv 0 \pmod{q}$. This is also not possible, since $1 \le j - i < j + i < q$, and p is invertible mod q. □

From this, it follows that we have

$$\prod_{i=1}^{Q} s_i = (-1)^{S(p,q)} Q!,$$

where $S(p, q)$ is the number of i between 1 and Q such that $s_i < 0$. If we put this together with the above, we find that $p^Q Q! \equiv (-1)^{S(p,q)} Q! \pmod{Q}$, so that $p^Q \equiv (-1)^{S(p,q)} \pmod{q}$. This gives the following:

Lemma 2.3.14

Let p and q be odd primes. Then,

$$\left(\frac{p}{q}\right) \equiv (-1)^{S(p,q)} \pmod{q}.$$

We therefore need to understand the parity of $S(p, q)$ and $S(q, p)$, and find a relationship between the two of them. More generally, if a is any odd number, let s_i be the unique integer between $-Q$ and Q such that $s_i \equiv ai \pmod{q}$, and let $S(a, q)$ be the number of i between q and Q such that $s_i < 0$.

Proposition 2.3.15

Let a be an odd positive integer, and let $T(a, q)$ denote the sum:

$$T(a,q) = \sum_{j=1}^{Q} \left\lfloor \frac{aj}{q} \right\rfloor.$$

Then, $S(a, q) \equiv T(a, q) \pmod{2}$.

Proof. For $1 \leq j \leq Q$, let r_j be the unique integer between 1 and q such that $r_j \equiv aj \pmod q$. We then have $aj = \lfloor \frac{aj}{q} \rfloor q + r_j$. Note that $\sum_{j=1}^{Q} j = \frac{Q(Q+1)}{2} = \frac{q^2-1}{8}$. On the other hand, we've seen that as j runs from 1 to Q, $|s_j|$ runs over the integers from 1 to Q. Thus, we have

$$\frac{q^2-1}{8} = \sum_{j=1}^{Q} j = \sum_{j=1}^{Q} |s_j| = \sum_{\substack{1 \leq j \leq Q \\ s_j > 0}} s_j - \sum_{\substack{1 \leq j \leq Q \\ s_j < 0}} s_j.$$

Note that if $s_j > 0$, then $r_j = s_j$. Otherwise, $r_j = s_j + q$, and the latter happens $S(a,q)$ times. Rewriting the last two sums in terms of the r_j then gives the following:

$$\frac{q^2-1}{8} = qS(a,q) + \sum_{j=1}^{Q} r_j - 2 \sum_{\substack{1 \leq j \leq Q \\ s_j < 0}} r_j.$$

Reducing modulo 2, and using that q is odd, we get

$$S(a,q) \equiv \frac{q^2-1}{8} + \sum_{j=1}^{Q} r_j \pmod 2.$$

On the other hand, we have

$$a\frac{q^2-1}{8} = \sum_{j=1}^{Q} aj = \sum_{j=1}^{Q} \left\lfloor \frac{aj}{q} \right\rfloor q + r_j = qT(a,q) + \sum_{j=1}^{Q} r_j.$$

Reducing modulo 2 and using that q, a are both odd, we find

$$T(a,q) \equiv \frac{q^2-1}{8} + \sum_{j=1}^{Q} r_j \pmod 2.$$

The result then follows. □

On the other hand, we have the following:

Proposition 2.3.16

Let p, q be odd primes. Then,

$$T(p,q) + T(q,p) = \frac{(p-1)(q-1)}{4}$$

Proof. Consider the set of pairs of integers (x, y) in the rectangle $1 \leq x \leq \frac{p-1}{2}$, $1 \leq y \leq \frac{q-1}{2}$. Clearly, there are $\frac{(p-1)(q-1)}{4}$ such pairs.

Now consider the line L given by $qx = py$. Since p, q are prime, there are no x, y in the rectangle that are on this line. For a given x, a point (x, y) lies below L if $y \leq \frac{qx}{p}$. Therefore, there are $\lfloor \frac{qx}{p} \rfloor$ points in the rectangle that lie below L and have x-coordinate x. Summing over all x, we attain $\sum_{j=1}^{\frac{p-1}{2}} \left\lfloor \frac{qx}{p} \right\rfloor$ points in the rectangle below L.This is precisely $T(q,p)$.

Similarly, for a given point y, (x, y) lies above L if $x \leq \frac{py}{q}$.Summing over all y, we find $T(p,q)$ points in the rectangle above L. The claim then follows. □

Theorem 2.3.17 (Law of Quadratic Reciprocity)

Let p and q be odd primes. Then, we have the following

- $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ if either p or q is 1 mod 4.
- $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$ if both p and q are 3 mod 4.

Equivalently, we have

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Proof. Let p, q be odd primes. Then,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{S(p,q)}(-1)^{S(q,p)} = (-1)^{T(p,q)}(-1)^{T(q,p)} = (-1)^{\frac{(p-1)(q-1)}{4}}.$$

□

§2.3.ii Gauss Sums

Here is another alternative proof of the law of quadratic reciprocity, one that uses Gauss sums.

Definition 2.3.18. Let p be a prime, and $\zeta \in \mathbb{C}$ be a p-th primitive root of unity. Then, the **Gauss sum** is the following sum:

$$\tau_P = \sum_{a \not\equiv 0 \pmod{p}} \left(\frac{a}{p}\right) \zeta^a \in \mathbb{Z}[\zeta],$$

where $\left(\frac{a}{p}\right)$ is the Jacobian symbol.

Lemma 2.3.19

Let p be a prime. Then,

$$\sum_{a \in F_p} \left(\frac{a}{p}\right) = \sum_{a \in F_p^\times} \left(\frac{a}{p}\right) = 0.$$

Proof. $\left(\frac{0}{p}\right) = 0$ by definition, and so the first sum is equal to the second sum. These are equal to the number of quadratic residues minus the number of quadratic non-residues. These are the same cardinality, so this is equal to 0. □

Theorem 2.3.20

Let p be a prime. We have

$$\tau_p^2 = \left(\frac{-1}{p}\right) p.$$

Proof. For this, we compute τ_p^2 two different ways. We have

$$\tau_p^2 = \sum_{a\in F_p^\times}\left(\frac{a}{p}\right)\zeta^a\cdot\sum_{b\in F_p^\times}\left(\frac{b}{p}\right)\zeta^b = \sum_{a,b\in F_p^\times}\left(\frac{a}{p}\right)\left(\frac{b}{p}\right)\zeta^{a+b}$$
$$= \sum_{c\in F_p^\times}\sum_{b\equiv ac \pmod p}\left(\frac{c}{p}\right)\zeta^{a+b} = \sum_{c\in \mathbb{F}_q^\times}\left(\frac{c}{p}\right)\cdot\sum_{a\in F_p^\times}\zeta^{a(1+c)}.$$

Note that in the above, we've taken c so that $b \equiv ac \pmod{()}p$. There are two cases to consider. If $c = -1$, then we have

$$\sum_{a\in F_p^\times}\zeta^{a(1+c)} = \sum_{a\in F_p^\times}\zeta^0 = |\mathbb{F}_p^\times| = p-1.$$

Next, if $c \neq -1$, then we note that ζ^{1+c} is again a p-th primitive root of unity. p is prime, so all p-th root of unities that aren't equal to 1 are primitive roots themselves. Define $\Phi_p(x) = \frac{x^p-1}{x-1}$. Then, η is a root of $\Phi_p(x)$ if and only if η is a primitive pth root of unity. Further, η is a root of x^p-1 if and only if η is a root of unity. Thus, we have

$$\prod_{a=1,\ldots,p-1}(x-\zeta^{(1+c)a}) = \Phi_p(x) = 1+x+\cdots+x^{p-1} = -1.$$

The claim then follows by the previous lemma. □

Theorem 2.3.21

Every elements of $\mathbb{Z}[\zeta]$ can be written uniquely as a sum

$$a_0 + a_1\zeta + \cdots + a_{p-2}\zeta^{p-2}, \quad a_i \in \mathbb{Z}.$$

Proof. By definition every element of $\mathbb{Z}[\zeta]$ can be written in the form $P(\zeta)$, where $P \in \mathbb{Z}[x]$. Let Φ_p denote the polynomial

$$\Phi_p(x) = \frac{x^p-1}{x-1} = 1+x+\cdots+x^{p-1}.$$

As Φ_p is monic we may apply long division to write $P(x)$ as

$$P(x) = Q(x)\Phi_p(x) + R(x), \quad Q(x), R(x) \in \mathbb{Z}[x]$$

such that $\deg(R(x)) < \deg(\Phi_p) = p-1$. Since $\Phi_p(\zeta) = 0$, we get that $P(\zeta) = R(\zeta)$. Now we only need to show uniqueness. If there were an element of $\mathbb{Z}[\zeta]$ which can be written in two different ways in the format above, then there is a $P(x) \in \mathbb{Z}[x]$ such that $\deg(P(x)) < p-1$ and $P(\zeta) = 0$. This means that $P(x)$ and $\Phi_p(x)$ have a common root, so their greatest common divisor in $\mathbb{Q}[x]$ is not a unit. Therefore $\Phi_p(x)$ is not irreducible in $\mathbb{Q}[x]$. So it will be sufficient to show that $\Phi_p(x)$ is irreducible in $\mathbb{Q}[x]$ in order to derive a contradiction.

Since $\Phi_p(x)$ is primitive, it will be enough to prove that $\Phi_p(x)$ is irreducible in $\mathbb{Z}[x]$. The same holds if and only if

$$\Phi_p(x+1) = (x+1)^{p-1} + \cdots + (x+1) + 1 = x^{p-1} + \cdots + p$$

is irreducible in $\mathbb{Z}[x]$. However the latter polynomial is Eisenstein, so it is irreducible. □

Definition 2.3.22. $\Phi_n(x)$ is called the ***n*-th cyclotomic polynomial**.

Remark 2.3.23 — As we can see in the proof above, we have that $\Phi_p(x)$ is irreducible, by Eisenstein.

Alternative Proof of the Law of Quadratic Reciprocity. We will compute the LHS in the quotient ring $R = \mathbb{Z}[\zeta]/(p)$. This is sufficient if we show that $1 \neq -1$ in R, which is equivalent to p not dividing 2 in R. If this were the case then there there is a $P(x) \in \mathbb{Z}[x]$ such that $\deg(P(x)) < p-1$ and $p \cdot P(\zeta) = 2$. But then $pP(x) - 2$ is also zero by the proposition above, and hence $p \mid 2$ in $\mathbb{Z}$, which is a contradiction.

As p is zero in R, we have $(a+b)^p = a^p + b^p$. Thus,

$$\tau_q^p = \sum_{a \in \mathbb{F}_q^\times} \left(\frac{a}{p}\right)^p \zeta^{ap} = \sum_{a \in \mathbb{F}_q^\times} \left(\frac{a}{p}\right) \zeta^{ap} = \left(\frac{p}{q}\right) \sum_{a \in \mathbb{F}_q^\times} \left(\frac{ap}{q}\right) \zeta^{ap} = \left(\frac{p}{q}\right) \tau_q.$$

Note that in the second equation we used that p is odd, that the Legendre symbol is a homomorphism in the third equation, and multiplication by p is a bijection on $\mathbb{F}_q^\times$ in the last equation. On the other hand,

$$\tau_p^q = (\tau_p^2)^{\frac{p-1}{2}} \tau_q = \left(\frac{-1}{p}\right)^{\frac{p-1}{2}} q^{\frac{p-1}{2}} \tau_q = (-1)^{\frac{(p-1)(q-1)}{4}} \left(\frac{q}{p}\right) \tau_q,$$

where we used the previous proposition in the second equation. We can compare these two equations to get

$$\left(\frac{p}{q}\right) \tau_q = (-1)^{\frac{(p-1)(q-1)}{4}} \left(\frac{q}{p}\right) \tau_q$$

in R. Noting that $\tau_q^2 = (\pm 1) \cdot q$, we get that q is invertible in $\mathbb{Z}/(p)$, so also in R. Thus, τ_q is invertible in R, and we get the desired result. $\square$

With this, together with Euler's criterion and Gauss's lemma, gives us a way of computing $\left(\frac{a}{p}\right)$ as long as we can factor a. Indeed, given a, we can assume that for $0 \leq a < p$, and write $a = \pm 2^r q_1^{k_1} \dots q_n^{k_n}$, with q_i odd primes less than p. Then, we have

$$\left(\frac{a}{p}\right) = \left(\frac{\pm 1}{p}\right) \left(\frac{2}{p}\right)^r \prod_i \left(\frac{q_i}{p}\right)^{k_i}.$$

Note that quadratic reciprocity implies that for each odd prime q, the question of whether q is a quadratic residue mod p has an answer in terms of congruence conditions mod q and mod 4. From this and the Chinese remainder theorem, we can deduce that the question: *for which primes p is a a quadratic residue mod p?* has an answer in terms of congruence conditions on p.

Example 2.3.24

We compute $\left(\frac{123}{97}\right)$. First, we note that

$$\left(\frac{123}{97}\right) = \left(\frac{26}{97}\right) = \left(\frac{2}{97}\right)\left(\frac{13}{97}\right).$$

$97 \equiv 1 \pmod 8$, so we get $\left(\frac{2}{97}\right) = 1$. Now for $\left(\frac{13}{97}\right)$, we apply the law of quadratic reciprocity.

$$\left(\frac{13}{97}\right) = \left(\frac{97}{13}\right) = \left(\frac{6}{13}\right) = \left(\frac{2}{13}\right)\left(\frac{3}{13}\right).$$

We get $\left(\frac{2}{13}\right) = -1$, and

$$\left(\frac{3}{13}\right) = \left(\frac{13}{3}\right) = \left(\frac{1}{3}\right) = 1.$$

Thus, we ultimately get $\left(\frac{123}{97}\right) = -1$.

More generally, one can ask, given a monic polynomial f with integer coefficients, for which primes p does f have a root? (The above case is the case of the polynomial $X^2 - a$.) Indeed, we are still extremely far from having a complete answer. One question that is natural to ask is: *for which f does above question have an answer given in terms of congruence conditions on p?* A deep branch of algebraic number theory called class field theory tells us that this will happen when the field extension determined by f has abelian Galois group. Beyond this we know very little, but there are connections to the theory of modular forms.

Definition 2.3.25. Let a, n be integers. We define the generalized **Legendre symbol** to be $\left(\frac{a}{n}\right)$, defined as follows:

$$\left(\frac{a}{n}\right) = \begin{cases} 1 & \text{if } (a,n) = 1 \text{ and } x^2 - a \text{ has a solution mod } n, \\ -1 & \text{if } (a,n) = 1 \text{ and } x^2 - a \text{ doesn't have a solution mod } n, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 2.3.26 — This is not to be confused with the Jacobi symbol. In times of ambiguity, we may refer to the generalized Legendre symbol by $\left(\frac{a}{n}\right)_L$ and the Jacobi symbol by $\left(\frac{a}{n}\right)_J$, when n is prime both of these functions agree so in this case there is no ambiguity.

Lemma 2.3.27

Let A be a commutative ring with unity and let $p \in A$ be

$$p * 1 = \underbrace{1 + 1 + \cdots + 1}_{p \text{ times}}$$

and let $a, b \in A$ such that $a \equiv b \pmod p$. Then, $a^{p^i} \equiv b^{p^i} \pmod{p^{i+1}}$, for all i.

Proof. For $i = 0$, this is part of the hypothesis of the theorem. Now, suppose we've shown that this is true for $i < n$, then i particular $a^{p^{n-1}} \equiv b^{p^{n-1}} \pmod{p^n}$. However,

$$a^{p^{n-1}} + r \cdot p^n = b^{p^{n-1}}$$

and raising to the pth power on both sides

$$a^{p^n} + \sum_{i=1}^{p} \binom{p}{i} r^i p^{ni} = b^{p^n}.$$

Since $p^{n+1} \mid p^{np}$ and $p \mid \binom{p}{i}$ for all i, we have the result. □

Lemma 2.3.28

Let p be an odd prime. The group $(\mathbb{Z}/p^k)^\times \cong \mathbb{Z}/(p-1) \times \mathbb{Z}/p^{k-1}$ as abelian groups.

Proof. We first look at the element $(1+p) \in \mathbb{Z}/p$. First, we note that since $(1+p) \equiv 1 \pmod{p}$. Then, by the previous lemma, we note that $(1+p^{p^{k-1}} \equiv 1 \pmod{p^k}$, so the order of $(1+p)$ in $(\mathbb{Z}/p)^\times$ divides p^{k-1}. On the other hand, we see that $\binom{p^i}{j}$ is divisible by at least $p^{p^i - m}$, where m is the largest integer such that p^m divides j. Finally, we see that

$$(1+p)^{p^i} = 1 + p^{i+1} + \sum_{l=2}^{p^i} p^l \binom{P^i}{l}.$$

Now note that if m_l is the largest integer such that p^{m_l} divides l, then for all l we have that $l - m_l \geq 2$ (this would be false if p is even). Therefore, $(1+p)^{p^i} \equiv 1 + p^{i+1} \pmod{p^{p^i+2}}$, for all i. Particularly, we see that this can only be 1 mod p^k when $i = k-1$, so $(1+p)$ has order exactly p^{k-1}.

Now note that the homomorphism of rings $\mathbb{Z}/p^k \to \mathbb{Z}/p$ is surjective; particularly, if $0 < a < p$ with $a \in \mathbb{Z}$, then $[a+p]_{p^k} \mapsto [a]_p$ under this map, and $[a+p]_{p^k}$ is a unit in $(\mathbb{Z}/p^k)^\times$, so this map also induces a surjective map of unites. The subgroup $(\mathbb{Z}/p^k)^\times$ consisting of those b such that $b \equiv 1 \pmod{p}$ consists of all elements $[1+np]_{p^k}$ with $0 \leq n < p^{k-1}$. This gives a subgroup of order p^{k-1} that is generated by $[1+p]_{p^k}$ by the computation in the previous paragraph. We choose $0 < a < p$ such that $[a]_p$ is a primitive root mod p, then $[a]_{p^k}^{p-1} \equiv 1 \pmod{p}$, and thus $[a]_{p^k}^{p-1} = (1+p)^i$ for some $0 < i \leq p^{k-1}$. Because $\langle [1+p]_{p^k} \rangle$ is a group of order p^{k-1} and $(p, p-1) = 1$, we have that $[1+p]_{p^k}^i$ has a $(p-1)$th root, i.e. there exists $[1+p]_{p^k}^j$ such that $[(1+p)]_{p^k}^{j(p-1)} = [a]_{p^k}^{p-1}$, whence $[a]_{p^k}[1+p]^{-j}$ is an element of order $(p-1)$. Because we have show that $(\mathbb{Z}/p^k)^\times$ contains two commuting elements of order $p-1$ and p^{k-1}, the result follows. □

Proposition 2.3.29

Let p be an odd prime and $k \geq 1$, $k \in \mathbb{Z}$, then $\left(\frac{a}{p^k}\right) = \left(\frac{a}{p}\right)$ for all $a \in \mathbb{Z}$, where $\left(\frac{a}{n}\right)$ denotes the Legendre symbol.

Proof. In the proof of the previous lemma we showed that $(\mathbb{Z}/p^k)^\times \cong (\mathbb{Z}/p)^\times \times \mathbb{Z}/p^{k-1}$ as abelian groups. We also showed that multiplication by $(p-1)$ is an isomorphism on

$\mathbb{Z}/p^{k-1}$, whence so is multiplication by 2. Thus an element x of the left hand side is a square if and only if $x \mapsto (x_1, x_2)$ where $x_1 \in (\mathbb{Z}/p)^\times$ is a square, and $x_2 \in \mathbb{Z}/p^{k-1}$ is arbitrary. But we note that $x_1 \equiv x \pmod p$, whence we derive the result. □

Theorem 2.3.30

Let $(a, n) = 1$, then $x^2 - a$ has a solution mod n if and only if $n = 2^m \cdot \prod_{i=1}^{k} p_k^{n_k}$ with the p_k odd and distinct and $\left(\frac{a}{p^k}\right)$ for all k, and $x^2 - a$ has a solution modulo 2^m. Again, $\left(\frac{a}{n}\right)$ denotes the Legendre symbol.

Proof. This follows from applying the Chinese remainder theorem and proposition 2.3.29 as the Chinese remainder theorem implies that an element in $(\mathbb{Z}/n)^\times$ is a square if and only if its image in each $(\mathbb{Z}/p_k^{n_k})^\times$ is a square and its image in $(\mathbb{Z}/2^m)^\times$ is a square. □

Remark 2.3.31 — Note that as a consequence the generalized Legendre symbol is not at all well-behaved in either the top or bottom argument when n is not prime power, it is not even multiplicative in the top argument in general as the equations $x^2 - 5, x^2 - 7, x^2 - 11$ all have no solutions modulo 12.

Remark 2.3.32 — There is a complement to this theorem 2.3.30 and proposition 2.3.29: an odd $a \in \mathbb{Z}$ is a quadratic residue modulo 2^n for $n \geq 3$ if and only if $a \equiv 1 \pmod 8$.

3 Using Norms

§3.1 Sum of Two Squares

Definition 3.1.1. We will say that a positive integer n is a **sum of two squares** if there exist integers x and y such that $n = x^2 + y^2$.

In a negative direction, note that any square is congruent to zero or one mod 4, so if n is a sum of two squares, then n cannot be congruent to 3 mod 4.

Lemma 3.1.2

If m and n are sums of two squares, then so is mn.

Proof. If $m = x^2 + y^2$ and $n = u^2 + v^2$, then

$$mn = (x^2 + y^2)(u^2 + v^2) = (xu - yv)^2 + (xv + yu)^2.$$

$\square$

Remark 3.1.3 — This is quite clear if we consider $x + yi, u + vi \in \mathbb{Z}[i]$, and use the multiplicativity of the norm.

Proposition 3.1.4 (Fermat Descent)

Suppose we have $a^2 + b^2 = pr$, with a, b integers and $1 < r < p$. Then there exists $1 \leq r' < r$ and integers x, y such that $x^2 + y^2 = pr'$.

Proof. Choose u, v such that $u \equiv a \pmod r$, $v \equiv b \pmod r$, and $-\frac{r}{2} \leq u, v \leq \frac{r}{2}$. Then, $u^2 + v^2 \equiv a^2 + b^2 \equiv 0 \pmod r$, so we can write $u^2 + v^2 = rr'$. Since u^2, v^2 are at most $\frac{r^2}{4}$, we have that $r' \leq \frac{r}{2}$. On the other hand, we cannot have $r' = 0$, as then $u = v = 0$, so $r \mid a$ and $r \mid b$, and so r^2 divides $a^2 + b^2 = pr$. It then follows that $r \mid p$, which is not possible. Therefore, $1 \leq r' < r$.

Now $(a^2 + b^2)(u^2 + v^2) = r'r^2p$. Rewriting this as a sum of two squares, we have the following:

$$(au + bv)^2 + (av - bu)^2 = r'r^2p.$$

Setting $x = \frac{au+bv}{r}$ and $y = \frac{av-bu}{r}$, then $x^2 + y^2 = r'p$. We need to check that x, y are integers, but

$$\begin{aligned} au + bv &\equiv a^2 + b^2 \equiv 0 \pmod r \\ av - bu &\equiv ab - ba \equiv 0 \pmod r, \end{aligned}$$

so this is clear. $\square$

Theorem 3.1.5 (Fermat's Two-Square Theorem)

Let p be a prime. Then, $x^2 + y^2 = p$ has a solution if and only if $p = 2$ or $p \equiv 1 \pmod 4$.

Proof. We first note that if p is a prime congruent to 1 mod 4, then Euler's criterion tells us that -1 is a quadratic residue mod p. We thus have an n, which we can take between 0 and $p-1$, such that $n^2 \equiv -1 \pmod p$. In particular, $p \mid n^2 + 1$, so we can write $n^2 + 1 = pr$, with $1 \le r < p$. Note that if $r = 1$ then we are done, as then $p = n^2 + 1$. For this, we use the technique of Fermat descent (proposition 3.1.4).

Now to express p as a sum of two squares, set $a_0 = n$, $b_0 = 1$, $r_0 = r$, and repeatedly apply the proposition to get a_i, b_i, r_i with $a_i^2 + b_i^2 = r_i p$ and r_i and the decreasing, but always at least one. Eventually one has $r_m = 1$ for some m and the claim follows. □

Notation 3.1.6. We denote by $\mathrm{ord}_p(n)$ to be the largest a such that p^a divides n.

Corollary 3.1.7

Let n be a positive integer, and suppose that for each prime p congruent to 3 mod 4, we have that $\mathrm{ord}_p(n)$ is even. Then n is the sum of two squares.

Proof. Such an n can be written as the product of a power of 2, powers of primes p congruent to 1 mod 4, and powers of q^2 for q congruent to 3 mod 4. But $2 = 1^2 + 1^2$, every p congruent to 1 mod 4 is a sum of two squares by the theorem, and $q^2 = q^2 + 0^2$ is a sum of two squares. Thus n is a product of sums of two squares and is therefore itself a sum of two squares. □

Remark 3.1.8 — It will turn out that these are precisely the n which can be expressed as the sum of two squares, but we will only prove this later, when we have more tools.

§3.2 Quadratic Integer Rings

Definition 3.2.1. Let $\alpha \in \mathbb{C}$. The ring $\mathbb{Z}[\alpha]$ is the smallest subring of $\mathbb{C}$ containing α.

As usual one has to check that this makes sense. An alternative definition is to take $\mathbb{Z}[\alpha]$ to be the intersection of all subrings of $\mathbb{C}$ containing α. This intersection clearly contains $0, 1, \alpha$, so it is nonempty. It's also closed under addition and multiplication, since it's an intersection of sets that are closed under these operations. Therefore it's a subring of $\mathbb{C}$ that, by construction is contained in any subring of $\mathbb{C}$ that contains α.

We now show that for $\alpha = i$, this agrees with the definition of the Gaussian integers $\mathbb{Z}[i]$. It is easy to see that the set of all integers of the form $a + bi$ is a subring of $\mathbb{C}$. On the other hand, any subring of $\mathbb{C}$ containing i is closed under addition and multiplication, and contains 1, so it contains every complex number of the form $a + bi$. Thus our new definition is consistent with our old one.

Note that for arbitrary α, it is not necessarily true that $\mathbb{Z}[\alpha]$ consists of all complex numbers of the form $a + b\alpha$ for $a, b \in \mathbb{Z}$ (although it always contains all complex numbers of that form). To see this, consider examples like $\mathbb{Z}[\pi]$ or $\mathbb{Z}[\frac{i}{p}]$ for p some prime, or $\mathbb{Z}[\beta]$ where β is a cube root of 2. For example, the last ring contains β^2, which is not of the form $a + b\beta$ for any integers a and b.

Definition 3.2.2. An element α of $\mathbb{C}$ is an **algebraic integer** of degree two (alternatively, a **quadratic algebraic integer**) if there exists a polynomial of the form $P(X) = X^2 + aX + b$ with a, b integers roots such that $P(X)$ has no rational roots and $P(\alpha) = 0$.

Example 3.2.3

i is an algebraic integer of degree two since $(i)^2 + 1 = 0$.

Suppose that α is an algebraic integer of degree two, and let a, b be integers such that $\alpha^2 + a\alpha + b = 0$. Then, for x, y, z, w integers, we have

$$(x + y\alpha)(z + w\alpha) = xz + (xw + yz)\alpha + yw\alpha^2 = (xa - byw) + (xw + yz - ayw)\alpha.$$

In particular the set of complex number of the form $x + y\alpha$ (x, y integers) is closed under addition and multiplication and is therefore a subring of $\mathbb{C}$. Since this subring contains α, it contains $\mathbb{Z}[\alpha]$. On the other hand it is clear that this subring is contained in $\mathbb{Z}[\alpha]$, so the two must be equal. We thus have the following proposition.

Proposition 3.2.4

If α is an algebraic integer of degree two, then $\mathbb{Z}[\alpha]$ is equal to the set of complex numbers of the form $x + y\alpha$, where x, y are integers.

Definition 3.2.5. For α an algebraic integer of degree two, we will say that $\mathbb{Z}[\alpha]$ is a **real quadratic subring** of $\mathbb{C}$ if α is a real number, and an **imaginary quadratic subring** of $\mathbb{C}$ if α is not real.

If $\mathbb{Z}[\alpha]$ is imaginary quadratic, then as with $\mathbb{Z}[i]$, we can define a *norm* $N \colon \mathbb{Z}[\alpha] \to \mathbb{Z}_{\geq 0}$ by setting $N(z) = z\overline{z}$. Note that $\overline{\alpha}$ is also a root of $\alpha^2 + a\alpha + b$ in this case, so we have

$$\alpha + \overline{\alpha} = -a, \quad \alpha\overline{\alpha} = b.$$

Therefore, we have

$$N(x + y\alpha) = (x + y\alpha)(x + y\overline{\alpha}) = x^2 + xy(\alpha + \overline{\alpha}) + y^2\alpha\overline{\alpha} = x^2 - axy + by^2.$$

Note that this norm is multiplicative: $N(zw) = N(z)N(w)$.

If $\mathbb{Z}[\alpha]$ is real quadratic, we let α^* denote the root of $X^2 + aX + b$ that is not equal to α (note that this is no longer equal to $\overline{\alpha}$). In this case, we define

$$N(x + y\alpha) = (x + y\alpha)(x + y\alpha^*) = x^2 - axy + by^2.$$

We therefore get a map $N \colon \mathbb{Z}[\alpha] \to \mathbb{Z}$. This is again multiplicative, but no long nonnegative.

Example 3.2.6

In $\mathbb{Z}[\sqrt{2}]$, $N(1+\sqrt{2}) = (1+\sqrt{2})(1-\sqrt{2}) = -1$.

We can study factorization in quadratic rings in a way analogous to the situation over $\mathbb{Z}$.

Definition 3.2.7. An element of $\mathbb{Z}[\alpha]$ is a **unit** if it has a multiplicative inverse; that is, if it lies in $\mathbb{Z}[\alpha]^\times$. Two elements z, w of $\mathbb{Z}[\alpha]$ are **associates** if there exists a unit $u \in \mathbb{Z}[\alpha]^\times$ such that $z = uw$.

Remark 3.2.8 — These are the same definitions for an arbitrary ring R.

Note that if $u \in \mathbb{Z}[\alpha]$ is a unit, there exists v such that $uv = 1$. Taking norms we find that$N(u) = \pm 1$ (if $\mathbb{Z}[\alpha]$ is imaginary quadratic, then this means $N(u) = 1$.) Conversely, if $N(u) = \pm 1$, then uu^*, so either u^* or $-u^*$ is a multiplicative inverse of u.

Example 3.2.9

The units in $\mathbb{Z}[i]$ are $\pm 1, \pm i$.

Definition 3.2.10. If z, u are elements of $\mathbb{Z}[\alpha]$, we say z **divides** u, writing $z \mid u$, if there exists $r \in \mathbb{Z}[\alpha]$ such that $u = zr$.

Remark 3.2.11 — Note that every element of a ring is divisible by units and its associates.

Definition 3.2.12. An element z of $\mathbb{Z}[\alpha]$ is **irreducible** if its only divisors are units and its associates.

Remark 3.2.13 — Note that for the ring $\mathbb{Z}$, we called such elements **prime**. Here the word irreducible is used instead, and save the word prime for a stronger condition (that is equivalent to irreducibility in the ring $\mathbb{Z}$).

Definition 3.2.14. Let z, w be elements of $\mathbb{Z}[\alpha]$, not both zero. An element r of $\mathbb{Z}[\alpha]$ is a **greatest common divisor** of z and w if

- $r \mid z$ and $r \mid w$
- Let $s \in \mathbb{Z}[\alpha]$. If $s \mid z$ and $s \mid w$, then $s \mid r$.

Remark 3.2.15 — Note that it is not clear from this definition that greatest common divisors always exist. They are also not unique: if a greatest common divisor does exist, then any associate is a greatest common divisor as well. On the other hand any two greatest common divisors are associates.

Proposition 3.2.16

Let z be an element of $\mathbb{Z}[\alpha]$. Then z factors into irreducibles.

Proof. Suppose there is an element of $\mathbb{Z}[\alpha]$ that does not admit a factorization into irreducibles. Then we can find such an element z such that $|N(z)|$ is minimal. Note that z itself cannot be irreducible, so we have $z = uv$ for elements u, v of $\mathbb{Z}[\alpha]$, neither of which is a unit. We have $|N(z)| = |N(u)||N(v)|$, with neither $|N(u)|$ or $|N(v)|$ equal to 1. Thus $|N(u)|$, $|N(v)| < |N(z)|$, so (by minimality) u and v admit factorizations into irreducibles. But then z does as well. $\square$

Example 3.2.17

On the other hand, it is not true, for an arbitrary $\mathbb{Z}[\alpha]$, that such factorizations are unique. For instance, in $\mathbb{Z}[\sqrt{-5}]$, one has

$$6 = (1 + \sqrt{-5})(1 - \sqrt{-5}) = 2 \cdot 3,$$

and it is not hard to see (for instance, by considering the norms of possible divisors) that $2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}$ are all irreducible, and none are associates of each other.

§3.3 Revisiting the Sums of Two Squares

Lemma 3.3.1

Let p be a prime in $\mathbb{Z}[i]$. Then there exists an integer prime q such that either $N(p) = q$ or $N(p) = q^2$. In the latter case p is an associate of q. Moreover, $N(p) = q$ if and only if q is the sum of two squares.

Proof. Let $n = N(p)$, and factor $n = q_1 \dots q_r$ as a product of integer primes. Since $p\overline{p}$, we have that $p \mid q_1 \dots q_r$, so (since p is prime) $p \mid q_i$ for some i. Let $q = q_i$, so $q = px$. Then, $q^2 = N(q) = N(p)N(x)$, so $N(x) \mid q^2$. Since $N(x) \neq 1$ (if it were p, would be a unit), we then have either $N(p) = q$ or $N(p) = q^2$. First suppose that $N(p) = q^2$. Since $p \mid q$, we have $pu = q$. Since $N(p) = N(q) = q^2$, we must have $N(u) = 1$, so u is a unit and p is an associate of q.

Next, suppose that $N(p) = q$. Writing $p = a + bi$, we see that $q = a^2 + b^2$. Conversely, if $q = a^2 + b^2$, then $q = (a + bi)(a - bi)$. Since $p \mid q$, we have either $p \mid a + bi$ or $p \mid a - bi$. In either case, $N(p) \mid q$ and therefore must equal q. $\square$

Remark 3.3.2 — Note that when $N(p) = q^2$, we have $p = \pm q$ or $\pm iq$ (def of associates).

Corollary 3.3.3

Primes of $\mathbb{Z}[i]$ are either of the form $a + bi$, where $a^2 + b^2$ is an integer prime, or q, where q is an integer prime that is not the sum of two squares.

We have thus reduced the whole question to the question of which primes are the sum of two squares, which we have already answered. However, our new perspective also gives us an alternative approach to this question. Particularly, we have the following proof of the two-square theorem via factorization in $\mathbb{Z}[i]$:

Alternative Proof of Fermat's Two Square Theorem(3.1.5). Let p be a prime congruent to 1 mod 4. As we have seen in our earlier proof of the two-square theorem, there exists n such that p divides n^2+1. Thus, in $\mathbb{Z}[i]$, we have that p divides $(n+i)(n-i)$. Since neither $\frac{n+i}{p}$ nor $\frac{n-i}{p}$ lie in $\mathbb{Z}[i]$, p doesn't divide $n+i$ or $n-i$ in $\mathbb{Z}[i]$. By the corollary p must be a sum of two squares. □

While this looks non-constructive, it isn't. If we want to find a, b such that $p = a^2 + b^2$, we use Euclid's algorithm to find a GCD of p and $n+i$. Since p doesn't divide either of $n+i$ or $n-i$, this GCD isn't a unit, nor is it an associate of p, so its norm is either 1 nor p^2. Thus, the GCD has norm exactly p. In fact, this is exactly what Fermat descent in the original proof is doing.

Since we've also shown that primes congruent to 3 mod 4 aren't the sum of two squares, we have the following complete characterization of sums of two squares.

Theorem 3.3.4

An integer n is a sum of two squares if and only if its prime factorization is of the following form:

$$n = 2^r p_1^{s_1} \dots p_k^{s_k} q_1^{t_1} \dots q_h^{t_h},$$

where p_i are primes congruent to 1 mod 4, the q_i are primes congruent to 3 mod 4 and the t_i are even.

Proof. Suppose n is of the form above, and write

$$z = (1+i)^r (a_1 + b_1 i)^{s_1} \dots (a_k + b_k i)^{s_k} q_1^{\frac{t_1}{2}} \dots q_h^{\frac{t_h}{2}},$$

where $p_i = N(a_i + b_i)$. Then $N(z) = n$. Conversely, if n is a sum of two squares, we can write $n = N(z)$, and factor

$$z = u(1+i)^r p_1^{s_1} \dots p_k^{s_k} q_1^{t_1} \dots q_h^{t_h}$$

in $\mathbb{Z}[i]$, where the p_i are primes in $\mathbb{Z}[i]$ whose norm is a prime congruent to 1 mod 4 and the q_i are primes in $\mathbb{Z}$ congruent to 3 mod 4. Then $N(z)$ has the claimed form. □

We can go further and compute the number of ways of representing n as a sum of two squares. Indeed, if we count the representations: $n = (\pm a)^2 + (\pm b)^2$ as being equivalent, then we are counting the number of ways to write $n = z\bar{z}$, with $z \in \mathbb{Z}[i]$, up to replacing z with an associate, or interchanging z and its conjugate. If we consider the factorization of n over $\mathbb{Z}$, and then further factoring over $\mathbb{Z}[i]$, assuming n can be written as a sum of squares at all, we can see that the number of in-equivalent ways to write n as a sum of two squares is given by:

$$\frac{1}{2}(s_1+1)(s_2+1)\dots(s_k+1)$$

if at least one s_i are odd, and

$$\frac{1}{2}\left[1 + (s_1+1)(s_2+1)\dots(s_k+1)\right]$$

if all s_i are even, where the s_i are the exponents of the primes congruent to 1 mod 4 in the factorization for n.

§3.4 Quadratics and Extensions of $\mathbb{Z}$

We can now generalize the approach we took in the new proof of the two square theorem to other Euclidean domains.

Definition 3.4.1. $\beta \in \mathbb{C}$ is a **quadratic number** if it is the root of a monic polynomial of degree 2 in $\mathbb{Z}[x]$.

Example 3.4.2

Take $d \in \mathbb{Z}$. Then, $\sqrt{d}$ is quadratic. It's a root of $x^2 - d \in \mathbb{Z}[x]$.

Lemma 3.4.3

If β is quadratic, then $\mathbb{Z}[\beta] = \{a + b\beta \mid a, b \in \mathbb{Z}\}$.

Proof. Obviously, every elements in the RHS is in the LHS, and β is in the RHS, so we just need to show that the RHS is a subring. We have $0 = 0 + 0 \cdot \beta$, and $1 = 1 + 0 \cdot \beta$, which is in the RHS, so we just need to see that this is closed under multiplication and addition. These are obvious:

$$(a + b\beta) + (a' + b'\beta) = (a + a') + (b + b')\beta$$
$$(a + b\beta)(a' + b'\beta) = (aa' + bb'\beta) + (ab' + a'b)\beta,$$

and so we can then plug in the quadratic that satisfies β^2 and regroup. □

Example 3.4.4

$\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain.

We can consider real β. If $\beta \in \mathbb{Q}$ and quadratic, then it follows that $\beta \in \mathbb{Z}$. Thus, $\mathbb{Z}[\beta] = \mathbb{Z}$. This follows by the following lemma:

Lemma 3.4.5 (Gauss's Lemma)

If $\beta \in \mathbb{Q}$ is the root of a monic polynomial in $\mathbb{Z}[x]$, then $\beta \in \mathbb{Z}$.

Proof. We write $\beta = \frac{a}{b}$, with $a, b \in \mathbb{Z}$ and $b \neq 0$. WLOG, assume that $(a, b) = 1$. Then, we have

$$\left(\frac{a}{b}\right)^n + \alpha_1 \left(\frac{a}{b}\right)^{n-1} + \cdots + \alpha_n = 0,$$

so

$$a^n + \alpha_1 a^{n-1} b + \cdots + \alpha_n b^n = 0,$$

so $b \mid a^n$, which is a contradiction unless $b = \pm 1$. □

Now consider the setting $x^2 + ax + b \in \mathbb{Z}[x]$, and let β, β' be its roots with $\beta \notin \mathbb{Z}$. Then, let $\beta' \notin \mathbb{Z}$ as well after factorization. Then, we get that $\mathbb{Z}[\beta] \subseteq \mathbb{Z}[\beta']$ and $\mathbb{Z}[\beta'] \subseteq \mathbb{Z}[\beta]$, so we actually get

$$\mathbb{Z}[\beta] = \mathbb{Z}[\beta'].$$

Now we define a map $()' \colon \mathbb{Z}[\beta] \to \mathbb{Z}[\beta]$ such that $a + b\beta \mapsto a + b\beta' \mapsto a + b\beta$. This is a ring homomorphism. We claim now that this map is a ring isomorphism; that is, it's an automorphism of $\mathbb{Z}[\beta]$. This respects $+$ (which is easy), and it respects $\cdot$, since β and β' satisfy the same monic quadratic.

Example 3.4.6

The map $\mathbb{Z}[i] \to \mathbb{Z}[i]$ that has root $i, -i$ sends $a + bi \mapsto a - bi$.

We can now define the norm in general. That is, for $a + b\beta \in \mathbb{Z}[\beta]$, we can let

$$N(a + b\beta) = (a + b\beta)(a + b\beta') = a^2 + 2b^2\beta\beta' + ab(\beta + \beta').$$

It's not always the case that this is nonnegative. For instance, if we let $\beta = \sqrt{d}$, with $d > 0$, then

$$(a + b\sqrt{d})(a - b\sqrt{d}) = a^2 - b^2 d < 0.$$

We can also construct the usual complex norm for $\mathbb{Z}[\beta]$. Taking $z \in \mathbb{Z}[\beta]$, we have $N(z) = a^2 + b^2\beta\overline{\beta} + ab(\beta + \overline{\beta})$, where $z = a + b\beta$. Then, considering the polynomial $(x - \beta)(x - \overline{\beta})$, we see that this polynomial must divide the monic polynomial $x^2 - cx - d$ that has β as a root. However, they're the same degree and both are monic, so this implies that

$$\beta + \overline{\beta} \in \mathbb{Z}, \quad \beta\overline{\beta} \in \mathbb{Z}.$$

Thus, our norm is multiplicative and if $x \mid y$ in $\mathbb{Z}[\beta]$, then $N(x) \mid N(y)$.

Example 3.4.7

$\mathbb{Z}[\sqrt{-5}]$ is not a UFD. This is because $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$.

We claim that $2, 3, 1 \pm \sqrt{-5}$ are all irreducible in $\mathbb{Z}[\sqrt{-5}]$. Let α be a unit. This is equivalent to $N(\alpha) = \pm 1$. This means that if any of $2, 3, 1 \pm \sqrt{-5}$, were reducible, it would be a factorization whose norm would be less than the norm of these elements, and these are also divisors. This implies that if any of these are reducible, then there exists an element of norm 2 or 3. However, this is not possible.

§3.4.i Primes and Quadratic Forms

Theorem 3.4.8

Ant element in $\mathbb{Z}[\alpha]$, where α is a quadratic algebraic integer, can be factored into irreducibles.

Proof. □

Remark 3.4.9 — The uniqueness fails in general. For instance, if we look at $\mathbb{Z}[\sqrt{-5}]$.

Definition 3.4.10. Let α be an imaginary algebraic integers, and b, c integers such that $\alpha^2 + b\alpha + c = 0$. We then refer to the polynomial $P(x, y)$ defined as follows as the **norm form** for α.

$$P(x, y) := N(x, y) = (x + y\alpha)(x + y\overline{\alpha}) = x^2 - bxy + cy^2.$$

Theorem 3.4.11

Suppose that unique factorization holds in $\mathbb{Z}[\alpha]$, and let p be an integer prime such that the polynomial $P(x,1) = x^2 - bx + c$ has a root mod p if and only if there exist integers x and y such that $P(x,y) = p$.

Proof. We have $x^2 - bx + c = (x+\alpha)(x+\overline{\alpha})$. Thus, if $x^2 - bx + c$ has a root mod p, then there exists x such that $p \mid (x+\alpha)(x-\overline{\alpha})$ in $\mathbb{Z}[\alpha]$. Since neither $\frac{x+\alpha}{p}$ nor $\frac{x-\alpha}{p}$ lies in $\mathbb{Z}[\alpha]$, p is not prime in $\mathbb{Z}[\alpha]$. Thus, p must be reducible in $\mathbb{Z}[\alpha]$, so $p = uv$ with neither u nor v unites, Then, $N(u)N(v) = p^2$, and neither $N(u)$ nor $N(v) = 1$, so $N(u) = p$ and the result follows.

Conversely, suppose there exist x and y such that $P(x,y) = p$. Then, $p \nmid y$. If $p \mid y$, then we'd have $x^2 - bxy + cy^2 = p$, implying $p \mid x$ as well. However, $x^2 - bxy + cy^2$ is divisible by p^2, so can't be equal to p. Therefore, y is invertible modulo p, and mod p we have $P(x,y) = 0$. However, modulo p, we have $P(x,y) = y^2 P(\frac{x}{y}, 1) = 0 \pmod{p}$, where $\frac{x}{y}$ means xy^{-1}. Thus, $\frac{x}{y}$ is a root of $P(x,1) = x^2 - bx + c \pmod{p}$. $\square$

Moreover, it is not hard to see that when p is odd, the polynomial $x^2 - bx + c$ has a root mod p if, and only if, the discriminant $b^2 - 4c$ is a square mod p:

Lemma 3.4.12

Let p be an odd prime. The polynomial $x^2 - bx + c$ has a root mod p if and only if $b^2 - 4c$ is zero or a quadratic residue mod p.

Proof. Suppose $b^2 - 4c \equiv d^2 \pmod{p}$, and let $x = 2^{-1}(b+d)$. then, $x^2 - bx + c = 4^{-1}(b^2 + 2bd + d^2 - 2b(b+d) + 4c) = 0$. Conversely, if x is a root of $x^2 - bx + c$, then $(2x-b)^2 = 4x^2 - 4bx + b^2 = b^2 - 4c$. $\square$

Since b and c are fixed, we can use quadratic reciprocity to turn this into a congruence condition on p.

For instance, if $\alpha = \frac{-1+\sqrt{-3}}{2}$, then $b = c = 1$ and $\mathbb{Z}[\alpha]$ is a Euclidean domain called the **Eisenstein integers**. Therefore, we see that any prime such that $x^2 - x + 1$ has a root mod p is of the form $x^2 + xy + y^2$, and conversely. On the other hand, the lemma shows that $x^2 - x + 1$ has a root mod p if and only if -3 is a zero or a quadratic residue mod p. Then, in particular, $x^2 - x + 1$ has a root mod p if and only if -3 is a square mod p. By quadratic reciprocity, this holds when $p = 3$ or $p \equiv 1 \pmod{3}$. Conversely, by directly checking mod 3, we see that $x^2 - xy + y^2$ is always 0 or 1 modulo 3, so the primes of the form $x^2 - xy + y^2$ are precisely 3 and the primes congruent to 1 mod 3.

Remark 3.4.13 — We note that

$$\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right] = \mathbb{Z}\left[\frac{-1+\sqrt{-3}}{2}\right],$$

if we consider these extensions as the smallest ring that consists each element and construct a bijection.

Remark 3.4.14 — We also have that

$$\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]^{\times} \cong \mathbb{Z}/6.$$

We state all the above in the theorem below:

Theorem 3.4.15

Let p be an odd prime, with the norm for equation $x^2+xy+y^2=p$ has an integer solution if and only if -3 is a quadratic residue modulo p if and only if $p=3$ or $p \equiv 1 \pmod 3$.

When unique factorization fails in $\mathbb{Z}[\alpha]$, the above theory fails. For instance, when $\alpha = \sqrt{-5}$ $(a=0, b=5)$, then $P(x) = x^2+5y^2$. In particular 3 is not of the form $P(x,y)$, even though x^2+5 has a root mod 3. In this situation the question of finding the primes of the form $P(x,y)$ has connections to class field theory and is a very deep part of modern algebraic number theory. For a taste of this, Cox's book Primes of the form x^2+ny^2 is good.

§3.5 Sums of Four Squares

§3.5.i Ring of Quaternions

Definition 3.5.1. The ring $\widetilde{H}$ of **quaternions** is the ring whose elements are formal sums $a+bi+cj+dk$, with $a,b,c,d \in \mathbb{R}$. Addition and multiplications are given as follows:

$$\begin{aligned}(a+bi+cj+dk)+(x+yi+zj+wk) &= (a+x)+(b+y)i+(c+z)j+(d+w)k \\ i^2 = j^2 &= k^2 = -1 \\ ij &= -ji = k \\ jk &= -kj = i \\ ki &= -ik = j\end{aligned}$$

Definition 3.5.2. A **skew field** (or division ring) is an algebraic structure with $1, 0, +, \cdot$ which satisfies all axioms of fields except commutativity for $\cdot$.

Let $z = a+bi+cj+dk$ be a quaternion. The conjugate z^* of z is the quaternion $a-bi-cj-dk$. Note that if z and w are quaternions, then $(zw)^* = w^*z^*$.

The norm $N(z)$ of z is given by

$$N(z) = zz^* = a^2+b^2+c^2+d^2.$$

Note that this is a real number, and hence commutes with all elements of $\widetilde{H}$. Thus we have

$$N(zw) = (zw)(zw)^* = zww^*z^* = zN(w)z^* = zz^*N(w) = N(z)N(w).$$

Proposition 3.5.3

$(\widetilde{H}, 1, 0, +, \cdot)$ is a skew field.

Proof. Let $0 \neq z \in \tilde{H}$. Then, $z \cdot z^* = a^2 + b^2 + c^2 + d^2 > 0$. Then,

$$z \cdot z^* \frac{1}{a^2 + b^2 + c^2 + d^2} = 1.$$

□

Remark 3.5.4 — $\mathbb{R}$ is the center of $\tilde{H}$. That is, the commuting elements of $\tilde{H}$ are $\mathbb{R}$.

Theorem 3.5.5

The only finite dimensional division algebras over $\mathbb{K}$ are $\mathbb{R}, \mathbb{C}, \tilde{H}$.

The norm provides an expression for the product of two sums of four squares as a sum of four squares:

$$\begin{aligned}(a^2 + b^2 + c^2 + d^2)(x^2 + y^2 + z^2 + w^2) &= N(a + bi + cj + dk)N(x + yi + zj + wk) \\ &= N((a + bi + cj + dk)(x + yi + zj + wk)).\end{aligned}$$

This final sum is equal to the following sum:

$$(ax - by - cz - dw)^2 + (ay + bx + cw - dz)^2 + (az - bw + cx + dy)^2 + (aw + bz - cy + dx)^2$$

In particular if m and n are integers that are representable as the sum of four integer squares, then so is their product mn. Thus to prove Lagrange's theorem it suffices to prove that every prime is the sum of four integer squares. Given that p is a prime, we know that if $p = 2$, or p is 1 mod 4, then we already know that p is a sum of two squares, hence also a sum of four squares. Therefore, it remains to prove that primes congruent to 3 mod 4 are sums of four squares. We will do this using a descent argument, similar to that used in the proof of the two square theorem.

Lemma 3.5.6

If $n, m \in \mathbb{N}$ are a sum of four squares, then so is $n \cdot m$.

Proof. Denote by $\mathbb{L}$ to be the **Lipschitz numbers**,

$$\mathbb{L} = \{a + bi + cj + dk \in \tilde{H} \mid a, b, c, d \in \mathbb{Z}\}.$$

Note that $\mathbb{L}$ is a subring. Thus, n is a sum of four squares if and only if n is the norm of a Lipschitz number. □

Remark 3.5.7 — If we use the ring of Hurwitz integers

$$\mathbb{H} = \{\frac{a}{2} + \frac{b}{2}i + \frac{c}{2}j + \frac{d}{2}k \mid a, b, c, d \in \mathbb{Z}, a \equiv b \equiv c \equiv d \pmod 2\},$$

we can use a Euclidean algorithm-type method. We will not do this here.

Lemma 3.5.8

Let p be a prime congruent to 3 mod 4. Then there exist x and y such that $x^2 + y^2 + 1 \equiv 0 \pmod{p}$.

Proof. It suffices to find an integer a such that a is a square mod p and $a+1$ is not. Since -1 is not a quadratic residue mod p we would then have $-a-1$ a quadratic residue mod p. Taking x such that $x^2 \equiv a \pmod{p}$ and y such that $y^2 \equiv -a-1 \pmod{p}$, the claim would follow. Suppose that we cannot do this. Then for each square a mod p, $a+1$ would also be a square mod p. In particular every congruence class mod p would be a square; since this doesn't happen we are done. $\square$

Fix a prime congruent to 3 mod 4. By the lemma we can find x and y such that $x^2 + y^2 = pr$ for some integer r. Since we only care about x and y mod p, we can further arrange that $|x|, |y| \leq \frac{p}{2}$. Then $r < p$. We are now ready to begin our descent:

Proposition 3.5.9

Suppose that for some $p > r > 1$, we have x, y, z, w such that $x^2 + y^2 + z^2 + w^2 = rp$. Then, there exist x', y', z', w' integers with $1 \leq r' < r$ and $(x')^2 + (y')^2 + (z')^2 + (w')^2 = r'p$.

Proof. There are two cases we must treat separately. First suppose that r is even. Then either all of x, y, z, w have the same parity, or two of them are odd and two are even. Permuting x, y, z, w as necessary, we can assume x and y have the same parity, as do z and w. Then set:

$$\begin{aligned} x' &= \frac{x+y}{2} \\ y' &= \frac{x-y}{2} \\ z' &= \frac{z+w}{2} \\ w' &= \frac{z-w}{2} \end{aligned}$$

It is then easy to verify that $(x')^2 + (y')^2 + (z')^2 + (w')^2 = \frac{r}{2}p$. Now suppose that r is odd. Select a, b, c, d such that $-\frac{r}{2} < a, b, c, d < \frac{r}{2}$ and $a \equiv x \pmod{r}$, $b \equiv y \pmod{r}$, $c \equiv z \pmod{r}$, and $d \equiv w \pmod{r}$. Note that we can get away with strict inequalities because r is odd. Now, since $x^2 + y^2 + z^2 + w^2 \equiv 0 \pmod{r}$, our congruences provide that $a^2 + b^2 + c^2 + d^2 \equiv 0 \pmod{r}$ as well. If we write $a^2 + b^2 + c^2 + d^2 = r'r$, and note that $r' < r$ since $a^2, b^2, c^2, d^2 < \frac{r^2}{4}$. On the other hand, r' is nonzero, for if $r' = 0$, then r divides all of x, y, z, w, and we'd have $r^2 \mid rp \implies r \mid p$. This can't happen, as $1 < r < p$, so we have $1 \leq r' < r$. We then have that

$$\begin{aligned} r'r^2p &= (a^2 + b^2 + c^2 + d^2)(x^2 + y^2 + z^2 + w^2) \\ &= (ax+by+cz+dw)^2 + (-ay+bx+cw-dz)^2 + (-az-bw+cz+dy)^2 + (-aw+bz-cy+dx)^2. \end{aligned}$$

Note that $ax + by + cz + dw \equiv x^2 + y^2 + z^2 + w^2 \equiv 0 \pmod{r}$. Similarly, each of the other terms are congruent to zero mod r. Therefore, we have the integers:

$$\begin{aligned} x' &= \frac{ax+by+cz+dw}{r} \\ y' &= \frac{-ay+bx+cw-dz}{r} \\ z' &= \frac{-az-bw+cx+dy}{r} \\ w' &= \frac{-aw+bz-cy+dx}{r} \end{aligned}$$

Then, we get $(x')^2+(y')^2+(z')^2+(w')^2 = r'p$, as desired.

To complete the proof, one begins with $x^2+y^2+1 = pr$ and repeatedly applies the proposition until $r = 1$. $\square$

Remark 3.5.10 — Note that in the above, when we find x', y', z', w', these are just the coordinates of $\alpha\overline{\beta}$, where

$$\alpha = a+bi+cj+dk, \beta = x+yi+zj+wk \in \tilde{H}.$$

Remark 3.5.11 — The descent in the proof of the two square theorem is really the Euclidean algorithm for $\mathbb{Z}[i]$ in disguise. This descent is also the Euclidean algorithm in a noncommutative setting. The associated ring is the ring of quaternions of the form $a+bi+cz+dw$, where either a, b, c, d are all integers or a, b, c, d are all half-integers (that is, fractions of the form $\frac{r}{2}$ with r odd).

§3.6 Sum of Three Squares

It's now natural to ask which positive integers are the sum of three integer squares. This turns out to be much more difficult. In one direction, you showed on an earlier example sheet that no integer of the form $4^t(8k+7)$ (t, k integers) is a sum of three squares. Conversely, one has the following theorem, which proof will be omitted due to it requiring more advanced methods.

Theorem 3.6.1
Every integer not of the form $4^t(8k+7)$ is a sum of three squares.

§3.7 Euclidean Domains

Definition 3.7.1. Let R be a commutative ring. Then, R is a **Euclidean domain** (or ED), if there exists a function $N\colon R \to \mathbb{N}$ such that

- $N(\alpha) = 0 \iff \alpha = 0$.
- $N(\alpha\beta) = N(\alpha)N(\beta)$.

- $\forall \alpha, \beta \in R$ with $\beta \neq 0$, there exists

$$\alpha = g \cdot \beta + r, \quad g, r \in R$$

such that $r = 0$ or $N(r) < N(\beta)$.

Example 3.7.2

If $R = \mathbb{Z}$ and $N = |\cdot|$, then this is a Euclidean domain.

Example 3.7.3

Let $R = k[x]$ for a field k, with $N = g^{\deg(\cdot)}$, $g > 1$.

Remark 3.7.4 — We have that $N(u) = 1$ if and only if u is a unit.

Theorem 3.7.5

Let R be a Euclidean domain, and let a and b be elements of R, with b nonzero. Then there exist elements x and y of r such that $xa + yb$ is a greatest common divisor of a and b.

Proof. The Euclidean algorithm gives an $a_r, x, y \in R$, such that $a_r = xa + yb$ and a_r divides both a and b. It suffices to show that a_r is a greatest common divisor of a and b. To see this, suppose that s divides a and b. Then s divides $xa + yb$, so $s \mid a_r$. Thus a_r has both properties required of a greatest common divisor. □

Theorem 3.7.6

Let $r, a, b \in R$, with R a Euclidean domain and r irreducible. Then if $r \mid ab$, either $r \mid a$ or $r \mid b$.

Proof. Suppose $r \nmid a$. Then the only common divisors of r and a are units, so we can write $1 = ax + ry$. Then $b = abx + rby$, and both abx and rby are divisible by r, so $r \mid b$. □

The above theorem amounts to the statement that irreducible elements of a Euclidean domain are prime. From this one can conclude, as one does over $\mathbb{Z}$, that any two factorizations of an element into irreducibles coincide up to permutation and replacing by associates.

The big theorem we want to show is that if R is a Euclidean domain, then we want to show that it has the fundamental theorem of arithmetic for R. That is, that ED $\implies$ UFD. To prove this though, discussion of ideals is necessary (because we need to talk about PIDs as well).

Lemma 3.7.7

Let R be an integral domain, and $a, b \in R$.

1. $a \mid b \iff (a) \supseteq (b)$.

2. a is associated to $b \iff (a) = (b)$.

3. a is irreducible $\iff (a)$ is maximal with respect to inclusion among proper principal ideals.

Proof. 1. $a \mid b \iff b = y \cdot a$, for $y \in R$. Equivalently, $b \in (a)$, so for $y \in R$, $yb \in (a)$. This is equivalent to saying that $(b) \subseteq (a)$.

The rest are exercises. □

Example 3.7.8

The Gaussian integers is a UFD.

Example 3.7.9

Let $R = \mathbb{Z}[x]$ is a UFD, but not a PID.

Theorem 3.7.10 (ED $\implies$ UFD)

Let R be a Euclidean domain and let $p_1 \ldots p_r = q_2 \ldots q_s = x$ be two factorizations of $x \in R$ into irreducibles. Then $s = r$ and one may rearrange the q's so that q_o is an associate of p_i for all i.

Example 3.7.11

$\mathbb{Z}$ is a Euclidean domain.

Lemma 3.7.12

$\mathbb{Z}[i]$ is a Euclidean domain with respected to N, the usual complex norm.

Proof. Let a and b be elements of $\mathbb{Z}[i]$ with b nonzero. Let $q' = \frac{a}{b}$; this is an element of $\mathbb{C}$, and we can write $q' = x + yi$, with x, y real. Set $q = c + di$, where c and d are integers such that

$$|x - c| \leq \frac{1}{2}, \quad |y - d| \leq \frac{1}{2}.$$

Then, $N(q - q') \leq \frac{1}{2}$. Set $r = a - bq$. Then, we have $N(\frac{a}{b} - q)N(b) \leq \frac{N(b)}{2}$, so $N(r) = N(a - bq) \leq \frac{N(b)}{2}$. Thus, N is a Euclidean norm and the claim follows. □

Corollary 3.7.13

$\mathbb{Z}[i]$ is a UFD.

On the other hand, $\mathbb{Z}[i]$ is not the only quadratic ring in which unique factorization holds. They seem to be relatively rare in general, however. For instance, there are only nine imaginary quadratic rings in which unique factorization holds: the rings $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$, $\mathbb{Z}[\frac{1+\sqrt{-11}}{2}]$, $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$, $\mathbb{Z}[\frac{1+\sqrt{-43}}{2}]$, $\mathbb{Z}[\frac{1+\sqrt{-67}}{2}]$, $\mathbb{Z}[\frac{1+\sqrt{-163}}{2}]$. It's easy to show this for the first five of these (they are Euclidean) and somewhat more difficult to show for the last four. The fact that these are the only imaginary quadratic rings where one has unique factorization was shown by Heegner.

§3.7.i UFDs

Definition 3.7.14. We define a binary relation $\sim$ on an integral domain R as follows. We say that a and b are **associates**, writing $a \sim b$, if and only if $a \mid b$ and $b \mid a$.

Theorem 3.7.15

Let R be an integral domain. Then, the following are equivalent.

1. R is a UFD
2. The following holds for R:
   - If $(a_1) \subseteq (a_2) \subseteq \cdots \subseteq (a_n) \subseteq \ldots$, where each $a_i \in R$ and eventually stationary.
   - Every irreducible element is prime.

Proof. ($1 \implies 2$): For all nonzero $a \in R$, define $d(a)$ as the number of irreducible factors in any factorization of a. Note that this is well defined.

$$a = u \cdot p_1 \ldots p_m,$$

so $d(a) = m$, if each p_i is prime. Note that $d(a) = 0$ if and only if $a \in R^\times$. If

$$a = v \cdot q_1 \ldots q_n,$$

then by unique factorization property, then $d(a) = n = m$. Now, we note the following about d.

- $a \mid b \implies d(a) \leq d(b)$.
- $a \mid b, d(a) = d(b) \implies a \sim b$.

Using this, we apply this to the sequence

$$(a_1) \subseteq (a_2) \subseteq \cdots \subseteq (a_n) \ldots,$$

with $a_2 \mid a_1, a_3 \mid a_2, \ldots, a_n \mid a_{n-1}$. This implies that

$$d(a_1) \geq d(a_2) \geq \cdots \geq d(a_n) \geq \ldots,$$

so this must become stationary at some point by non-negativity. Next, we get that from its stationary point, say from n, we have that $a_n \sim a_{n+1} \sim \cdots \sim a_{n+m} \sim \dots$. Ultimately, we combine these to get

$$(a_n) = (a_{n+1}) = \cdots = (a_{n+m}) = \dots$$

Consider now $x \mid ab$ that is irreducible. Then, there exists c, so we have $ab = xc$. Writing $a = u \cdot p_1 \dots p_m$, $b = v \cdot q_1 \dots q_n$, and $c = w \cdot v_1 \dots v_n$, we have

$$xc = wx \cdot v_1 \dots v_n = ab.$$

If $x \mid p_i$, then $x \mid p_i \mid a$, so true. On the other hand, if $x \mid q_i$, then $x \mid q_i \mid b$, so true. Thus, x is prime.

$(2 \implies 1)$: We use the ascending chain condition for principal ideals. Note that we can factor all elements into irreducibles. We can say $a = a_1$, and if it's irreducible, we're done. If note, we can write $a_1 = a_2 b_2$, where a_2, b_2 aren't units. If these aren't irreducible, then we're done. If note, we can continue by setting $a_2 = a_3 b_3$. This way we have

$$\cdots \mid a_n \mid \cdots \mid a_3 \mid a_2 \mid a_1,$$

such that

$$\cdots \not\sim a_n \not\sim \cdots \not\sim a_3 \not\sim a_2 \not\sim a_1,$$

This is equivalent to having

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots \subsetneq (a_n) \subsetneq \dots,$$

which isn't stationary, so contradiction. Then, we can find uniqueness by writing two representations for $a \in R$, and verify that two irreducibles (that are prime by hypothesis) are associated in the two representations. □

Proposition 3.7.16

The ascending chain condition holds in PIDs.

Proposition 3.7.17

Let R be a PID. Then, all irreducible elements in R are prime.

Proof. Let $p \mid ab$. Then,

$$(p) \subseteq (p, a) := \{px + ay \mid x, y \in R\} = \bigcap_{\substack{I \triangleleft R \\ a, b \in I}} I.$$

Then, the LHS is smaller than the RHS, and the LHS is an ideal, so we get equality. Now note that if p is irreducible, then its ideal (p) is maximal among proper principal ideals. However, in R, all ideal are principals, so (p) is a maximal ideal. Therefore, $(p) = (p, a)$ or $(p) = R$. In the first case, we have that

$$(p) = (p, a) \implies a \in (p) \implies a = px, x \in R \implies p \mid a.$$

In the second case, we have

$$(p, a) = R \implies \exists x, y \in R, ax + yp = 1 \implies abx + ybp = b \implies p \mid b.$$

Therefore, p is prime. □

Theorem 3.7.18

All PIDs are UFDs.

Proof. By the two results above, we get the result. ☐

Example 3.7.19

Consider $k[x] \subset k[x^{\frac{1}{2}}] \subset k[x^{\frac{1}{4}}] \subset \cdots \subset k[x^{\frac{1}{2^n}}]$ Then, all finitely generated ideal is prime, but the ring is not PID.

§3.7.ii UFDs for Integer Polynomials

Example 3.7.20

Consider $\mathbb{Z}[x]$. This is a UFD, but not a PID. For this, we consider the following ideal:

$$(2, X) = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in \mathbb{Z}, 2 \mid a_0\}.$$

This is not principal. To prove this, assume that $(2, X) = (p(x))$. Then, $p(x) \mid 2$, so $p(x) = c \in \mathbb{Z}$. Therefore, $c \mid x$, so $c \cdot g(x) = x$. Thus, $\frac{x}{c} \in \mathbb{Z}$, so $c = \pm 1 \notin (2, X)$.

Definition 3.7.21. A polynomial $p(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ is **primitive** if $(a_0, a_1, \ldots, a_n) = 1$ and if $b \mid a_i$ for all i, then b is a unit.

Lemma 3.7.22 (Gauss's Lemma)

Let $f(x), g(x) \in \mathbb{Z}[x]$ be primitive. Then, $f(x)g(x)$ is also a primitive.

Proof. It suffices to show that there exists a prime p that divides all coefficients of $f(x)g(x)$. We can consider the surjective ring homomorphism

$$\begin{aligned} \mathbb{Z}[x] &\to \mathbb{F}_p[x] \\ \sum a_n x^n &\mapsto \sum [a_n]_p x^n. \end{aligned}$$

Note that these are indeed ring homomorphisms, since the map

$$\begin{aligned} \mathbb{Z} &\to \mathbb{F}_p \\ a &\mapsto [a_n]_p \end{aligned}$$

is a ring homomorphism. Then, writing $f \cdot g = \sum c_n x^n$, with $p \mid c_i$ for all i is equivalent to saying that $f \cdot g \mapsto 0$ with respect to the homomorphism defined above. Thus, f, g being primitive implies that $f, g \not\equiv 0 \pmod{p}$, and so $f \cdot g \not\equiv 0$ in $\mathbb{F}_p$. We note that it isn't possible for $f \cdot g \equiv 0 \pmod{p}$ by a degree argument. Writing $\overline{f} = f \pmod{p}, \overline{g} = g \pmod{p}, \overline{fg} = fg \pmod{p}$, we have

$$\deg(\overline{f}) + \deg(\overline{g}) = \deg(\overline{fg}) = 0,$$

so contradiction. ☐

Note that if R is a PID and $(a_1, \ldots, a_n) = (b)$, then $b \sim (a_1, \ldots, a_n)$, the greatest common divisor of $a_1, \ldots, a_n$.

Now, how can we show that $\mathbb{Z}[x]$ is a UFD? We have $\mathbb{Z}[x] \subset \mathbb{Q}[x]$, which is a PID, so $\mathbb{Q}[x]$ is also a UFD. Now, for some nonzero $f \in \mathbb{Z}[x]$, we have that $f(x)$ is equal to the product of the g.c.d. of the coefficients of f multiplied by some primitive polynomial. Formally, we have

$$f(x) = \sum a_n x^n, \quad d = (a_0, \ldots, a_n),$$

and so

$$f(x) = \sum d a_n' x^n = d \underbrace{\sum a_n' x^n}_{\text{primitive}}.$$

This factorization is unique up to sign.

Lemma 3.7.23

Let $0 \neq f(x) \in \mathbb{Z}[x]$. Then, $f(x)$ is irreducible if and only if either

1. $\deg(f) = 0$ and $f = p \in \mathbb{Z}$ is a prime
2. $\deg(f) > 0$, and f is primitive, and f is irreducible as an element of $\mathbb{Q}[x]$.

Proof. □

Theorem 3.7.24

$\mathbb{Z}[x]$ is a UFD.

Proof. □

§3.7.iii Cubics

We now apply the previous theory of the inclusion of rings (ED $\implies$ PID $\implies$ UFD) to cubics. Consider the Diophantine equation

$$y^2 + 1 = x^3.$$

It has the trivial solution $(x, y) = (1, 0)$.

Theorem 3.7.25

The only solution of the Diophantine equation above has only one unique solution pair.

Proof. We shall factor this Diophantine equation in the ring $\mathbb{Z}[i]$.

$$(y + i)(y - i) = x^3.$$

Now, if d is a common divisor of $y + i$ and $y - i$, then $d \mid 2i \implies d \mid 2$. We now claim that $(y + i, y - i) = 1$. Notice that x must be odd. To see this, note that if $2 \mid x$, then

$8 \mid x^3$, and so we consider quadratic residues mod 8. Then, since $y^2 \equiv 0, 1 \pmod 8$ in all cases, x can't be even. The reason why we did this is because

$$x^3 = N(y+i) = N(y-i),$$

so the primes $\pm 1 \pm i \nmid y \pm i$. Thus, the only prime factors of d are those $\{\pm 1 \pm i\}$, so d has no prime factors, implying that $d \sim 1$. Therefore, $(y+i, y-i) = 1$. Writing $x = u \cdot p_1^{\alpha_1} \dots p_k^{\alpha_k}$, then

$$x^3 = u^3 \cdot p_1^{3\alpha_1} \dots p_k^{3\alpha_k} = (y+i)(y-i).$$

Now,

$$\begin{aligned} p_i \mid y+i &\implies p_i \nmid y-i \\ p_i \mid y-i &\implies p_i \nmid y+i. \end{aligned}$$

This implies that $y + i = v \cdot p_{i_1}^{3\alpha_{i_1}} \dots p_{i_k}^{3\alpha_{i_K}}$, where $v \in \mathbb{Z}[i]^\times$. We now claim that v is a cube in $\mathbb{Z}[i]$. In $\mathbb{Z}/4$, residue by mod 3 is a bijection as $(3, 4) = 1$. For any unit in $\mathbb{Z}[i]$ a cube, this implies $v = w^3, w \in \mathbb{Z}[i]$. Therefore,

$$y + i = w^3 \cdot z^3 = t^3, \quad t \in \mathbb{Z}[i].$$

This means that $y + i = (a + ib)^3$, where $a, b \in \mathbb{Z}$. We compute the RHS:

$$(a+ib)^3 = (a^3 - 3ab^2) + i(3a^2b - b^3).$$

We can then attain that $a = 0, b = -1$, so $y + i = i$. Thus, we get a unique solution. $\square$

Let $\beta \in \mathbb{C}$. Then, define $\mathbb{Z}[\beta]$ as the smallest subring of $\mathbb{C}$ containing β. This is well defined, since this set can be expressed as the following

$$\bigcap_{\substack{R \subset \mathbb{C} \\ \text{is a subring} \\ \beta \in R}} R,$$

and the subrings is a subring.

4 Approximation

§4.1 Pell's Equation

Definition 4.1.1. Let $d > 0$ be a non-square integer. Then, the following equation is called **Pell's equation**.

$$x^2 - dy^2 = 1.$$

Note that $\mathbb{Z}[\sqrt{d}]$ is a real quadratic subring of $\mathbb{C}$, and its norm form is provided by

$$N(x + y\sqrt{d}) = x^2 - dy^2.$$

Therefore, Pell's equation is equivalent to finding elements that have norm 1 in $\mathbb{Z}[\sqrt{d}]$. Such elements are units, and the norm is multiplicative, these elements form a subgroup $\mathbb{Z}[\sqrt{d}]^{\times,1}$, of $\mathbb{Z}[\sqrt{d}]^{\times}$. Two obvious such elements are ± 1. All other such elements are of the form $x + y\sqrt{d}$, with $x, y \in \mathbb{Z}$ and $y \neq 0$.

Lemma 4.1.2

Let $x + y\sqrt{d}$ be an element of $\mathbb{Z}[\sqrt{d}]^{\times,1}$. Then,

- We have $x, y > 0$ if and only if $x + y\sqrt{d} > 1$.
- We have $x > 0$, $y < 0$ if and only if $0 < x + y\sqrt{d} < 1$.
- We have $x < 0$, $y > 0$ if and only if $-1 < x + y\sqrt{d} < 0$.
- We have $x, y < 0$ if and only if $x + y\sqrt{d} < -1$.

Proof. It is clear that if $x, y > 0$ then $x + y\sqrt{d} > 1$. But then $x - y\sqrt{d} = (x + y\sqrt{d})^{-1}$ lies between 0 and -1. Similarly, $-x + y\sqrt{d}$ lies between -1 and 0, and $-x - y\sqrt{d}$ is less than -1. Thus we have the rightward implication in each of the four claims above. But since the four cases are mutually exclusive and exhaust all the possibilities, the leftward implications hold as well. □

Lemma 4.1.3

Let $z = x + y\sqrt{d}$, $z' = x' + y'\sqrt{d}$ be two elements of $\mathbb{Z}[\sqrt{d}]^{\times,1}$ with x, y, x', y' all positive. Then $z > z'$ if and only if $y > y'$.

Proof. We have $z - \frac{1}{z} = x + y\sqrt{d} - (x - y\sqrt{d}) = 2y\sqrt{d}$. Since $z - \frac{1}{z}$ is an increasing function for z positive, we have $z > z'$ if and only if $z - \frac{1}{z} > z' - \frac{1}{z'}$ if and only if $y > y'$. □

Suppose we have a nontrivial element of $\mathbb{Z}[\sqrt{d}]^{\times,1}$ (that is, one other than ± 1). Without loss of generality we can take x and y positive. There then exists $\epsilon = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]^{\times,1}$ such that x and y are both positive and y is as small as possible. We will call ϵ the **fundamental 1-unit** in $\mathbb{Z}[\sqrt{d}]$. By the previous two lemmas it is the smallest element of $\mathbb{Z}[\sqrt{d}]^{\times}$ that is greater than one.

Proposition 4.1.4

Suppose there exists a nontrivial element of $\mathbb{Z}[\sqrt{d}]^{\times,1}$. Then, every element of $\mathbb{Z}[\sqrt{d}]^{\times,1}$ is of the form $\pm\epsilon^n$, for some n in $\mathbb{Z}$, where ϵ is the fundamental 1-unit.

Proof. Let z be an element of $\mathbb{Z}[\sqrt{d}]^{\times,1}$. After negating z and/or replacing z by $\frac{1}{z}$, as necessary, we can assume $z > 1$. Since ϵ is also greater than one, there exists n such that $\epsilon^n \leq z < \epsilon^{n+1}$. Then $\epsilon^{-n}z$ is a 1-unit with $1 \leq \epsilon^{-n}z < \epsilon$. Since ϵ is the smallest 1-unit greater than one we have $\epsilon^{-n}z = 1$. □

Remark 4.1.5 — Note that this proposition is equivalent to saying that

$$\mathbb{Z}[\sqrt{d}]^{\times} = \{\pm 1\} \oplus \epsilon^{\mathbb{Z}}.$$

Note that $\epsilon^{\mathbb{Z}}$ is free. This is because $\epsilon^n > \epsilon^{n-1}$, and so the order of ϵ is ∞.

From here, we show that there are always elements of $\mathbb{Z}[\sqrt{d}]^{\times,1}$ other than ± 1, so that there are infinitely many solutions to Pell's equation. The key idea is to consider the equation in the rationals; that is, to consider when $\frac{x}{y}$ is, in some sense, as close as possible to $\sqrt{d}$. This notion suggest that we try to approximate $\sqrt{d}$ with the rationals. It's clear that if p and q are integers, and α is an irrational, then we can make $|\frac{p}{q} - \alpha|$ as small as is desired. However, we may need to make q large to do so. Therefore, we'll be interested in approximations to α where the error $|\frac{p}{q} - \alpha|$ is small compared to $\frac{1}{q^n}$, for various n (in other words, in the p-adic sense). As n gets bigger, it becomes difficult. Note that when $n = 1$, it's easy: For any α any any q, there exists a p such that

$$|\frac{p}{q} - \alpha| < \frac{1}{q}.$$

Here is the case for $n = 2$, which is provided by Dirichlet:

Theorem 4.1.6

Let α be an irrational number, and $Q > 1$ an integer. Then, there exist p, q integers, with $1 \leq q < Q$, such that $|p - q\alpha| < \frac{1}{Q}$.

Proof. For $1 \leq k \leq Q - 1$, let $a_k = \lfloor k\alpha \rfloor$, so that $0 < k\alpha - a_k < 1$. Now partition the interval $[0, 1]$ into Q sub-intervals of length $\frac{1}{Q}$. Then, one of these intervals contains two elements of the set

$$\{0, \alpha - 1, 2\alpha - a_2, \ldots, (Q-1)\alpha - a_{Q-1}, 1\}.$$

The difference between these two elements is of the form $p - q\alpha$, where p and q are integers and $q < Q$, and this difference is less than $\frac{1}{Q}$. □

Corollary 4.1.7

For any irrational α, there are infinitely many $\frac{p}{q}$ such that $|\alpha - \frac{p}{q}| < \frac{1}{q^2}$.

Proof. It suffices to show, given any $\frac{p}{q}$ with $|\alpha - \frac{p}{q}| < \frac{1}{q^2}$, that we can find another $\frac{p'}{q'}$, with $|\alpha - \frac{p'}{q'}| < \frac{1}{(q')^2} < |\alpha - \frac{p}{q}|$. Suppose, given such a $\frac{p}{q}$, and choose Q such that $\frac{1}{Q} < |\alpha - \frac{p}{q}|$. Then, by the theorem, there exist p', q' with $q' < Q$, and $|\alpha - \frac{p'}{q'}| < \frac{1}{Qq'} < \frac{1}{(q')^2}$. The claim then follows. $\square$

Theorem 4.1.8

For any non-square d, there is a nontrivial solution to $x^2 - dy^2 = 1$.

Proof. The corollary provides infinitely many pairs (p_i, q_i) such that $|p_i - q_i\sqrt{d}| < \frac{1}{q_i}$. Note that then $|p_i + q_i\sqrt{d}| < \frac{1}{q_i} + 2q_i\sqrt{d} < 3q_i\sqrt{d}$. We thus have

$$|N(p_i - q_i\sqrt{d})| = |(p_i - q_i\sqrt{d})(p_i + q_i\sqrt{d})| < 3\sqrt{d}.$$

Thus for some M between $-3\sqrt{d}$ and $3\sqrt{d}$, there are infinitely many pairs (p_i, q_i) such that $N(p_i + q_i\sqrt{d}) = M$. Since there are finitely many congruence classes mod M, there is some pair (p_0, q_0) such that there are infinitely many pairs (p_i, q_i) with $N(p_i + q_i\sqrt{d}) = M$, $p_i \equiv p_0 \pmod{M}$, and $q_i \equiv q_0 \pmod{M}$. Then, for (p_i, q_i) and (p_j, q_j) any two such pairs, consider the quotient:

$$\frac{p_i - q_i\sqrt{d}}{p_j - q_j\sqrt{d}} = \frac{(p_ip_j - dq_iq_j) + (p_iq_j - p_jq_i)\sqrt{d}}{M}.$$

The congruence conditions show that this quotient lies in $\mathbb{Z}[\sqrt{d}]$, and it has norm 1 by multiplicativity of the norm. $\square$

§4.1.i Varying Pell's Equation

Note that we can also apply these techniques to solving

$$x^2 - dy^2 = -1.$$

Solutions correspond to elements of norm -1 in $\mathbb{Z}[\sqrt{d}]$; if $x + y\sqrt{d}$ is one solution, the others are given by $\pm(x + y\sqrt{d})\epsilon^n$, where ϵ is the fundamental 1-unit. Note, however, that unlike for 1 units there may be no -1-units at all (consider, for instance, $d = 3$, where the equation has no solutions mod 3 and thus no integer solutions.)

§4.2 Liouville's Theorem

We now ask whether we can find infinitely many $\frac{p}{q}$ such that

$$|\frac{p}{q} - \alpha| < \frac{1}{q^e},$$

for some $e > 2$. It turns out that there for many irrational α, the answer is no. In particular, let's make the following definition:

Definition 4.2.1. Let d be a positive integer. A complex number α is **algebraic** of degree d if there is a degree d (not necessarily monic) polynomial $P(x)$, with integer coefficients, such that $P(\alpha) = 0$, and no such polynomial of degree less than d. α is **transcendental** otherwise.

Theorem 4.2.2 (Liouville's Theorem)

Let α be a real number that is algebraic of degree d. Then, for any real number $e > d$, there are at most finitely many rational number $\frac{p}{q}$ such that $|\frac{p}{q} - \alpha| < \frac{1}{q^e}$.

Proof. Let $P(x)$ be a polynomial of degree d, with integer coefficients, such that $P(\alpha) = 0$. Choose ϵ such that $P(x)$ has no roots other than α on the closer interval $[\alpha - \epsilon, \alpha + \epsilon]$. Write $P(x) = (x - \alpha)Q(x)$, where $Q(x)$ is a monic polynomial with real coefficients of degree $d - 1$. Since $|Q(x)|$ is a continuous, real valued function, there is a real number $K > 0$ such that $|Q(x)| \leq K$ on the compact set $[\alpha - \epsilon, \alpha + \epsilon]$. Now, suppose that we have $\frac{p}{q}$ with $|\frac{p}{q} - \alpha| < \frac{1}{q^e}$. There are only finitely many q such that $\frac{1}{q^e} \geq \epsilon$, so we may assume that $\frac{1}{q^e} < \epsilon$. Now, on one hand we have

$$|P(\frac{p}{q})| = |(\frac{p}{q} - \alpha)||Q(\frac{p}{q})| < \frac{1}{q^e}K.$$

On the other hand, since P has degree d and integer coefficients, the denominator of $P(\frac{p}{q})$, when written in lowest terms, is a divisor of q^d. However, $\frac{p}{q}$ is not a root of $P(x)$, so $P(\frac{p}{q})$ is nonzero and hence $|P(\frac{p}{q})| \geq \frac{1}{q^d}$. Putting these inequalities together, we get

$$\frac{1}{q^d} \leq |P(\frac{p}{q})| < \frac{1}{q^e}K.$$

Rewriting, we find that $q^{e-d} < K$. Since $e - d > 0$, there are only finitely many q for which this is possible. □

Example 4.2.3

Let us find all pairs of positive integers (p, q) such that $|\sqrt[3]{2} - \frac{p}{q}| < \frac{1}{q^4}$.

Solution. We trace the proof of Liouville's theorem. Note that $\sqrt[3]{2}$ satisfies $p(x) = x^3 - 2$. Suppose that $\frac{p}{q}$ satisfies $|\sqrt[3]{2} - \frac{p}{q}| < \frac{1}{q^4}$. We can then write:

$$|p(\frac{p}{q}) - 2| = |(\frac{p}{q})^3 - 2| = |\sqrt[3]{2} - \frac{p}{q}| \cdot |(\frac{p}{q})^2 + (\frac{p}{q})\sqrt[3]{2} + \sqrt[3]{4}|$$

By the irrationality of $\sqrt[3]{2}$, the LHS is at least $\frac{1}{q^3}$. Thus, if we have a constant K with $|(\frac{p}{q})^2 + (\frac{p}{q})\sqrt[3]{2} + \sqrt[3]{4}| \leq K$, then we can attain that

$$\frac{1}{q^4} > |\sqrt[3]{2} - \frac{p}{q}| \geq \frac{1}{Kq^3}, \tag{$*$}$$

so that $q < K$. We now find this K. Suppose that $q \geq 2$. Then,

$$\frac{p}{q} < \sqrt[3]{2} + \frac{1}{2^4} < \frac{3}{2} + \frac{1}{2^4} = \frac{25}{2^4} < \frac{3}{2}.$$

Then, we have that

$$|(\frac{p}{q})^2 + (\frac{p}{q})\sqrt[3]{2} + \sqrt[3]{4}| < \frac{9}{4} + \frac{3}{2} \cdot \frac{3}{2} + \frac{9}{4} = \frac{27}{4} < 7.$$

Thus, let us take $K = 7$ for simplicity.We now case work for different q using ($*$). If $q = 1$, we attain that $p = 1, 2$. Now, we consider $q \in \{2, 3, \ldots, 6\}$, but we can rule all these cases out. For instance, try $q = 2$. Then, it is clear that the closest to values inside the bracket, 1 and $\frac{3}{2}$ lies more than $\frac{1}{16}$ away from $\sqrt[3]{2}$ (which is approximately $1.25992\ldots$). Note without this approximation at the end, it gets quite tricky. □

The set of polynomials $P(x)$ with integer coefficients is countable. Since each such polynomial has finitely many roots the set of algebraic numbers is countable. Since the set of reals is uncountable this means that, in a very strong sense, almost every real number is transcendental. In spite of this it is very hard to give an example of a single real number that is provably transcendental. (In fact, e and π are examples of transcendental numbers, but this is much harder than what we'll do.) Liouville's theorem gives one approach to proving that a given number is transcendental: show that it admits too many good rational approximations. If we can show that for any e, there exist infinitely many $\frac{p}{q}$ such that $|\frac{p}{q} - \alpha| < \frac{1}{q^e}$, then Liouville's theorem tells us that α can't be algebraic of any degree, and hence must be transcendental.

Example 4.2.4

Define **Liouville's number**, which is the real number α by

$$\alpha = \sum_{n=1}^{\infty} \frac{1}{10^{n!}}.$$

This clearly converges. We can find rational approximations to α simply by truncating the series. For each k, let α_k be the sum

$$\alpha_k = \sum_{n=1}^{k} \frac{1}{10^{n!}}.$$

On one hand, α_k is rational with denominator $10^{k!}$. On the other hand, we have

$$|\alpha - \alpha_k| = \sum_{n=k+1}^{\infty} \frac{1}{10^{n!}} < \frac{2}{10^{(k+1)!}}.$$

Fix a positive integer d. For any $k > d$, we have $\frac{2}{10^{(k+1)!}} < \frac{1}{(10^{k!})^d}$. Thus, there are infinitely many k such that $|\alpha - \alpha_k| < \frac{1}{(10^{k!})^d}$. Liouville's theorem thus tells us that α cannot be algebraic of degree d. Since d was arbitrary, α must be transcendental.

Liouville's theorem tells us that algebraic numbers are difficult to approximate well by rationals. In fact, they are even more difficult to approximate than Liouville's theorem would suggest. For instance, there's Roth's theorem:

Theorem 4.2.5 (Roth's Theorem)

Suppose α is algebraic. Then, for any $\epsilon > 0$, there are only finitely many rational numbers $\frac{p}{q}$ such that $|\frac{p}{q} - \alpha| < \frac{1}{q^{2+\epsilon}}$.

In other words, for algebraic numbers, you cannot do any better than Dirichlet's theorem. Roth's theorem is considerably harder to prove than Liouville's, and we will not give a proof here. Note that the stronger bound gives proofs that additional real numbers are transcendental; for instance, one can prove with Roth's theorem that the number

$$\beta = \sum_{n=1}^{\infty} \frac{1}{10^{3n}}$$

is transcendental, which isn't possible with Liouville's theorem.

§4.3 Continued Fractions

§4.3.i Rationals

Given a rational number $\frac{p}{q}$, we can write $\frac{p}{q} = a_0 + r_0$, where a_0 is an integer and $0 \le r_0 < 1$ is between 0 and 1. In other words, we write

$$\frac{p}{q} = \underbrace{\left\lfloor \frac{p}{q} \right\rfloor}_{a_0} + \underbrace{\left\{ \frac{p}{q} \right\}}_{r_0}$$

If r_0 is not zero, then we can write $\frac{1}{r_0} = a_1 + r_1$, with a_1 and integer and $0 \le r_1 < 1$. We then have:

$$\frac{p}{q} = a_0 + r_0 = a_r + \frac{1}{a_1 + r_1}.$$

If we continue like this, as long as r_i is nonzero, we set

$$\frac{1}{r_i} = a_{i+1} + r_{i+1},$$

with a_i an integer and $0 \le r_{i+1} < 1$. The denominators of the r_i are strictly decreasing, so eventually som $r_n = 0$ and we have

$$\frac{p}{q} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \cdots + \frac{1}{a_n}}}.$$

This expression is called the **continued fraction expansion** of $\frac{p}{q}$. It is closely related to the Euclidean algorithm. Indeed, it's not difficult to see that the a_i are the quotients q_i from the algorithm applied to the pair $\frac{p}{q}$. Note that each a_i is an integer and for $i \ge 1$, we have $a_i \ge 1$.

§4.3.ii Infinite Continued Fractions

Now let α be an irrational real number. As above, we can write $\alpha = a_0 + r_0$, where a_0 is an integer and $0 \le r_0 < 1$, and then for each i set

$$\frac{1}{r_i} = a_{i+1} + r_{i+1},$$

with a_{i+1} an integer and $0 \le r_{i+1} < 1$. Note that unlike in the rational case, this sequence will never terminate, as r_i is always irrational and thus never zero. We write:

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \ldots}}$$

and call this the **continued fraction expansion** of α. We will now unravel what this exactly means, but first, we introduce some notation.

Notation 4.3.1. For $a_0, \ldots, a_n$ real numbers, we define $[a_0; a_1, \ldots, a_n]$ to be the real number:

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_n}}}$$

when this expression is well-defined (that is, when it does not involve a division by zero.)

Definition 4.3.2. We say that a continued fraction is **almost** or **eventually periodic**, if

$$[a_0; a_1, a_2, \dots, a_n, \dots] = \lim_{n\to\infty} [a_0; a_1, \dots, a_n],$$

if the limit on the right hand side exists.

The value of this expression can computed by a recurrence relation, as is in the following lemma.

Lemma 4.3.3

Given $a_0, \dots, a_n$ real numbers, define p_i, q_i for $0 \le i \le n$ by $p_0 = a_0$, $q_0 = 1$, $p_1 = a_0a_1 + 1$, $q_1 = a_1$, together with the recurrences:

$$p_i = a_i p_{i-1} + p_{i-2}$$
$$q_i = a_i q_{i-1} + q_{i-2}$$

Then, $[a_0; a_1, \dots, a_n] = \frac{p_n}{q_n}$, assuming no q_i is zero.

Proof. We prove this by induction on n; the cases $n = 0$ and $n = 1$ are clear. Let p_i' and q_i' be the numbers defined by the recurrence attached to the sequence $a_0, a_1, \dots, a_{n-2}, a_{n-1} + \frac{1}{a_n}$. The induction hypothesis tells us that $[a_0; a_1, \dots, a_{n-2}, a_{n-1} + \frac{1}{a_n}] = \frac{p'_{n-1}}{q'_{n-1}}$. By the recurrence defining the p' and q', we have

$$\frac{p'_{n-1}}{q'_{n-1}} = \frac{(a_{n-1} + \frac{1}{a_n})p'_{n-2} + p'_{n-3}}{(a_{n-1} + \frac{1}{a_n})q'_{n-2} + q'_{n-3}}$$

The sequences defining p_i', q_i' and p_i, q_i agree for $i \le n-2$, the latter is equal to:

$$\frac{(a_{n-1} + \frac{1}{a_n})p_{n-2} + p_{n-3}}{(a_{n-1} + \frac{1}{a_n})q_{n-2} + q_{n-3}}$$

Multiplying both numerator and denominator by a_n, and using $p_{n-1} = a_{n-1}p_{n-2} + p_{n-3}, p_n = a_np_{n-1} + p_{n-2}$ (and similarly for q_n) we find that this expression is equal to $\frac{p_n}{q_n}$. □

Note that if $a_i \ge 1$ for all i (as will be the case, for instance, if the a_i come from taking a continued fraction expansion of some real number), then the q_i are all nonzero and form a strictly increasing sequence. (Indeed, if all a_i are at least one, then $q_i \ge q_{i-1} + q_{i-2} \ge 2q_{i-2}$, so this sequence increases exponentially fast.) From this, we can attain that $q_n \ge 2^{\lfloor \frac{n}{2} \rfloor}$.

Example 4.3.4

We compute $\sqrt{2}$. We have that

$$\sqrt{2} = 1 + (\sqrt{2} - 1).$$

Then, we have that $\frac{1}{\sqrt{2}-1} = \sqrt{2} + 1 = 2 + (\sqrt{2} - 1)$. Therefore, we've shown that

$$\sqrt{2} = 1 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \dots}}}$$

Example 4.3.5

Next we compute $\sqrt{3}$. We have

$$\sqrt{3} = 1 + (\sqrt{3} - 1).$$

Then,

$$\frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2} = 1 + \frac{\sqrt{3}-1}{2}.$$

Then,

$$\frac{2}{\sqrt{3}-1} = \sqrt{3} + 1 = 2 + (\sqrt{3} - 1).$$

Therefore, we get

$$\sqrt{3} = 1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}$$

Example 4.3.6

Next, we compute $\sqrt{13}$. Thus, we have

$$\begin{aligned}
\sqrt{13} &= 3 + (\sqrt{13} - 3) \\
\frac{1}{\sqrt{13}-3} &= \frac{\sqrt{13}+3}{4} = 1 + \frac{\sqrt{13}-1}{4} \\
\frac{4}{\sqrt{13}-1} &= \frac{\sqrt{13}+1}{3} = 1 + \frac{\sqrt{13}-2}{3} \\
\frac{3}{\sqrt{13}-2} &= \frac{\sqrt{13}+2}{3} = 1 + \frac{\sqrt{13}-1}{3} \\
\frac{3}{\sqrt{13}-1} &= \frac{\sqrt{13}+1}{4} = 1 + \frac{\sqrt{13}-3}{4} \\
\frac{4}{\sqrt{13}-3} &= \sqrt{13} + 3 = 6 + (\sqrt{13} - 3).
\end{aligned}$$

Therefore, $\sqrt{13} = [3; 1, 1, 1, 1, 6]$. We apply this to the equation $x^2 - 13y^2 = 1$, and we take x to be the numerator and y to be the denominator of each index. We have

$$\begin{aligned}
3^2 - 13 &= -4 \\
4^2 - 13 &= 3 \\
7^2 - 13 \cdot 2^2 &= -3 \\
11^2 - 13 \cdot 3^2 &= 4 \\
18^2 - 13 \cdot 5^2 &= -1.
\end{aligned}$$

However, we can just take the positive square root, as this should equal 1 then:

$$(18 + \sqrt{13} \cdot 5)^2 = 649 + 180 \cdot \sqrt{13}.$$

Thus, we get $649^2 - 13 \cdot 180^2 = 1$.

Definition 4.3.7. Now suppose we have an infinite sequence $a_0, a_1, a_2, \dots$ of real numbers, and assume that $a_i \geq 1$ for $i \geq 1$. Define p_i and q_i by the recurrence given above. We call $\frac{p_i}{q_i}$ the i**th convergent** to the continued fraction:

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}}.$$

Lemma 4.3.8

We have $p_n q_{n-1} - q_n p_{n-1} = (-1)^{n-1}$.

Proof. This is again by induction on n; the base case $n = 1$ is clear. Assume this is true for $n - 1$. Then by the defining recurrence for the p_i and q_i , we have:

$$\begin{aligned} p_n q_{n-1} - q_n p_{n-1} &= (a_n p_{n-1} + p_{n-2}) q_{n-1} - (a_n q_{n-1} + q_{n-2}) p_{n-1} \\ &= p_{n-2} q_{n-1} - q_{n-2} p_{n-1} \\ &= -(-1)^{n-2}, \end{aligned}$$

and the claim follows. □

It follows that $|\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}}| = \frac{1}{q_n q_{n-1}} < \frac{1}{q_{n-1}^2} < \frac{1}{2^n}$. Since the q_n increase exponentially, it follows that the $\frac{p_i}{q_i}$ form a Cauchy sequence, and hence converge to a real number.

Now, a natural question to ask at this point is: "if the sequence $a_0, a_1, \dots$ arises from the continued fraction expansion of an irrational number α, is the limit of the convergents of $\frac{p_n}{q_n}$ equal to α?". This is indeed the case:

Lemma 4.3.9

Let α be an irrational real number, and let $a_0, a_1, \dots.$ be the sequence of integers arising from its continued fraction expansion. Let $\frac{p_n}{q_n}$ be the nth convergent. Then $\frac{p_n}{q_n} < \alpha$ if n is even, and $\frac{p_n}{q_n} > \alpha$ if n is odd.

Proof. Once again we use induction on n; the case $n = 0$ is clear. Note that $[a_1; a_2, \dots, a_n]$ is the $(n-1)$st convergent to $\frac{1}{\alpha - a_0}$. Thus, by the induction hypothesis, if n is odd we have

$$[a_1; a_2, \dots, a_n] < \frac{1}{\alpha - a_0}.$$

Therefore,

$$\alpha < a_0 + \frac{1}{[a_1; a_2, \dots, a_n]} = [a_0; a_1, a_2, \dots, a_n].$$

If n is even the same argument works with the inequalities reversed. □

Corollary 4.3.10

Let α be an irrational real number, and let $a_0, a_1, \dots$ be the sequence of integers arising from its continued fraction expansion. Let $\frac{p_n}{q_n}$ be the nth convergent. Then $|\alpha - \frac{p_n}{q_n}| < \frac{1}{q_n q_{n+1}}$. In particular the limit $\frac{p_n}{q_n}$ as $n \to \infty$ is α.

Proof. Since exactly one of $\frac{p_n}{q_n}$ and $\frac{p_{n+1}}{q_{n+1}}$ is less than α, we have

$$|\alpha - \frac{p_n}{q_n}| < |\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n}| = \frac{1}{q_n q_{n+1}}$$

Since the q_n increase exponentially quickly the second claim is clear. $\square$

Note that since $\frac{1}{q_n q_{n+1}} < \frac{1}{q_n^2}$, this gives a second, completely constructive proof of Dirichlet's theorem on rational approximations.

Remark 4.3.11 — From this, we get that an irrational α has an eventually periodic continued fraction expansion if and only if α is a quadratic irrational.

There is a sense in which the convergents to the continued fraction expansion of α are the best possible rational approximations to α. We make this precise. Fix α real and irrational, and as above define a_i and r_i by setting $\alpha = a_0 + r_0$, with a_0 an integer and $0 \leq r_0 < 1$, and $\frac{1}{r_i} = a_{i+1} + r_{i+1}$ with a_{i+1} an integer and $0 \leq r_{i+1} < 1$. Define p_n and q_n from the sequence $a_0, a_1, \ldots$ by the recurrences of the previous section.

Lemma 4.3.12

For all n, we have:

$$\alpha = \frac{p_n + p_{n-1} r_n}{q_n + q_{n-1} r_n}$$

Proof. For $n = 1$, this is easy. Assume it is true for $n - 1$; that is, that we have:

$$\alpha = \frac{p_{n-1} + p_{n-2} r_{n-1}}{q_{n-1} + q_{n-2} r_{n-1}}$$

We have $\frac{1}{r_{n-1}} = a_n + r_n$. Substituting $\frac{1}{a_n + r_n}$ for r_{n-1} in the above and using the relations $p_n = a_n p_{n-1} + p_{n-2}$, $q_n = a_n q_{n-1} + q_{n-2}$ yields the claimed result. $\square$

Corollary 4.3.13

For all n, $|\alpha - \frac{p_n}{q_n}| < |\alpha - \frac{p_{n-1}}{q_{n-1}}|$.

Proof. We have:

$$\frac{|\alpha - \frac{p_n}{q_n}|}{|\alpha - \frac{p_{n-1}}{q_{n-1}}|} = \frac{q_{n-1}}{q_n} |\frac{q_n \alpha - p_n}{q_{n-1}\alpha - p_{n-1}}|$$

Substituting $\alpha = \frac{p_n + p_{n-1} r_n}{q_n + q_{n-1} r_n}$, we find that the right hand side is equal to

$$\frac{q_{n-1}}{q_n} |\frac{q_n \frac{p_n + p_{n-1} r_n}{q_n + q_{n-1} r_n} - p_n}{q_{n-1} \frac{p_{n-1} + p_{n-2} r_{n-1}}{q_{n-1} + q_{n-2} r_{n-1}} - p_{n-1}}|.$$

Simplifying, we find that this is equal to the following

$$\frac{q_{n-1}}{q_n} |\frac{q_n p_{n-1} r_n - p_n q_{n-1} r_n}{q_{n-1} p_n - p_{n-1} q_n}| = \frac{q_{n-1}}{q_n} r_n < 1.$$

The claim then follows. $\square$

Theorem 4.3.14

Let h, k be integers with $|k| \le q_n$. Then $|\frac{h}{k} - \alpha| \ge |\frac{p_n}{q_n} - \alpha|$. In other words, $\frac{p_n}{q_n}$ is the best rational approximation to α with denominator at most q_n.

Proof. Suppose that $|\frac{h}{k} - \alpha| < |\frac{p_n}{q_n} - \alpha|$. Then by the corollary we also have $|\frac{h}{k} - \alpha| < |\frac{p_{n-1}}{q_{n-1}} - \alpha|$. Suppose that n is even. Then we have

$$\frac{p_n}{q_n} < \alpha < \frac{p_{n-1}}{q_{n-1}},$$

and our inequalities imply

$$\frac{p_n}{q_n} < \frac{h}{k} < \frac{p_{n-1}}{q_{n-1}}.$$

If n is odd the same holds with the inequalities reversed, by a similar argument. Either way we have

$$|\frac{h}{k} - \frac{p_{n-1}}{q_{n-1}}| < |\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}}| = \frac{1}{q_n q_{n-1}}.$$

On the other hand, the left hand side of this is a positive rational number with denominator kq_{n-1}, so must be at least $\frac{1}{kq_{n-1}}$. This is a contradiction since $k \le q_n$. □

Theorem 4.3.15

Let d be a non-square integer. Then, for all solutions to Pell's equation $x^2 - dy^2 = 1$, it appears as an approximation $[a_0; a_1, \dots, a_n]$ of the continued fraction expansion $[a_0; a_1, \dots, a_n, \dots]$ for some n. The first approximation with this property is the fundamental solution.

Remark 4.3.16 — Note that there are two methods to solve general recurrence: realizing that we get a basis for a vector space vs generating functions.

§4.3.iii Möbius Transforms

Definition 4.3.17. We define the group $G = GL_2(\mathbb{Z})$ to be the set of invertible 2×2 matrices with integer entries, i.e. the set of matrices

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

such that $a, b, c, d \in \mathbb{Z}$ and $\det(T) = ad - bc = \pm 1$, with identity the identity matrix I_2, multiplication coming from matrix multiplication, and inverse

$$T^{-1} = \det(T) \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}.$$

This group G is interesting because it acts on the space $\mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$ in the following way.

Definition 4.3.18. Let $z \in \mathbb{P}^1(\mathbb{C})$, if $z \in \mathbb{C}$ and $g \in G$ then

$$g \cdot z := \frac{az+b}{cz+d},$$

where if $cz + d = 0$ we take $g \cdot z = \infty$. And $g \cdot \infty = \infty$ if $c = 0$ and $g \cdot \infty = \frac{a}{c}$ otherwise. The maps induced on $\mathbb{P}^1(\mathbb{C})$ by the action of G are called **Möbius transformations**.

Remark 4.3.19 — The space $\mathbb{P}^1(\mathbb{C})$ can actually be made into a topological space, by insisting that the map $\frac{1}{z}$ is continuous, and that $\mathbb{C} \subset \mathbb{P}^1(\mathbb{C}$ has the subspace topology. With this topology the action of G on $\mathbb{P}^1(\mathbb{C}$ is continuous, and even a map of Riemann surfaces.

Because the entries of the matrix $g \in G$ are real numbers, and in fact integers, we see that the action of G restricts to an action on the subspaces $\mathbb{Q}_* := \mathbb{Q} \cup \{\infty\}$ and $\mathbb{R}_* := \mathbb{R} \cup \{\infty\}$. It turns out the dynamics of the action of G on $\mathbb{R}_*$ are quite intimately related to the behavior of continued fractions.

Definition 4.3.20. We define the special Möbius transformations to be those induced by the elements $T_a := \begin{pmatrix} 0 & 1 \\ 1 & a \end{pmatrix}, S_a := \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \in G$.

Proposition 4.3.21

Let $\alpha = [a_0; a_1, \ldots, a_n, \ldots]$ be a positive real number. Then the convergents $\frac{p_n}{q_n} := [a_0; a_1, \ldots, a_n]$ are obtained by applying the Möbius transformation $(S_{a_0} \circ Ta_1 \circ \cdots \circ T_{a_n})$ to the element 0. Further there is an equality of Matrices

$$\begin{pmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{pmatrix} = (S_{a_0} \circ Ta_1 \circ \cdots \circ T_{a_n})$$

Proof. The first statement of course follows from the second by applying both matrices to 0, so we only prove the second statement. This, as always, can be proved by induction. When $n = 1$ we are asking if

$$\begin{pmatrix} a_0 & 1 + a_0 a_1 \\ 1 & a_1 \end{pmatrix} = S_{a_0} \circ T_{a_1} = \begin{pmatrix} 1 & a_0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix},$$

which follows easily. . On the other hand, we have

$$\begin{aligned} \begin{pmatrix} p_{n-1} & a_n(p_{n-1}) + p_{n-2} \\ q_{n-1} & a_n(q_{n-1}) + q_{n-2} \end{pmatrix} \cdot T_{a_n}^{-1} &= \begin{pmatrix} p_{n-1} & a_n(p_{n-1}) + p_{n-2} \\ q_{n-1} & a_n(q_{n-1}) + q_{n-2} \end{pmatrix} \cdot \begin{pmatrix} -a_n & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} p_{n-2} & p_{n-1} \\ q_{n-2} & q_{n-1} \end{pmatrix}. \end{aligned}$$

and by the inductive hypothesis we have the result. □

As a result of this, we can reprove many of our results from class. For instance our definition of the p_n, q_n, could have been as the entries of the vector $(S_{a_0} \circ Ta_1 \circ \cdots \circ T_{a_n}) \circ \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and the recurrence relation defining p_n, q_n would follow easily. In addition our result on the relationship $p_n q_{n-1} - q_n p_{n-1} = (-1)^{n-1}$ can be interpreted in the following way.

Corollary 4.3.22

$$p_{n-1}q_n - p_n q_{n-1} = \det\begin{pmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{pmatrix} = (-1)^n.$$

Finally we can come back to our study of quadratic irrationalities via continued fractions.

Lemma 4.3.23

Let $\alpha = \frac{1}{c}(a + b\sqrt{d})$ with $d \geq 0$ and $a, b, c \in \mathbb{Z}$ with a, b, c not having any prime factors in common amongst the three of them then:

1. The number α satisfies a quadratic polynomial with integer coefficients, and every real solution to such a polynomial is of this form.

2. If α is such a number, then so is $g \cdot \alpha$ for any $g \in G$.

Proof. (1): We note that one direction is just the quadratic equation. In the other direction, we note that $p(x) = x^2 + \gamma x + \tau$ has α as a root if and only if $\gamma = \frac{-2a}{c}$, $\tau = \frac{a^2 - db^2}{c^2}$, whence we see that

$$q(x) = c^2x^2 - 2acx + (a^2 - db^2)$$

is the desired polynomial.

(2): we note that α satisfies a quadratic polynomial $\tau x^2 + \gamma x + \theta$ then $\alpha = g^{-1} \cdot g(\cdot\alpha)$, and if we define $\beta = g \cdot \alpha$ then substituting we have

$$\tau\left(\frac{d\beta - c}{a\beta - b}\right) \pm \gamma\left(\frac{d\beta - c}{a\beta - b}\right) + \theta = 0.$$

clearing denominators we solve for a quadratic polynomial of degree at most 2 satisfied by β as desired. □

Remark 4.3.24 — We note that if α satisfies an irreducible quadratic polynomial it must be irrational, in which case we see that $g \cdot \alpha$ is also irrational, because the action of G on $\mathbb{P}^1(\mathbb{C})$ preserves $\mathbb{Q}_*$. As such we see by pure thought that in this case if the leading term τ is nonzero, then so is the leading term of the polynomials satisfied by $g \cdot \alpha$.

Theorem 4.3.25

If α is a real number with an eventually repeating continued fraction expansion, then α is a quadratic irrational or a rational number.

Proof. As usual, without loss of generality we take $\alpha > 0$. If α's continued fraction expansion terminates we know that α is rational and vice versa, so without loss of generality we assume α is irrational. By the remark we see that the action of G preserves the quadratic irrationals, so by replacing α by $(T_{x_n} \circ \cdots \circ T_{x_1}) \cdot \alpha$ for suitable x_i we may assume α is purely periodic, whence it is a fixed point of some $T = (T_{a_1} \circ \cdots \circ T_{a_k}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We can also verify that $T \neq \pm I_2$ because $T_{a_k}^{-1}$ and $-T_{a_k}^{-1}$ have negative coefficients, whereas $(T_{a_1} \circ \cdots \circ T_{a_{k-1}})$ has only positive coefficients. So finally we have $T \cdot \alpha = \alpha$ whence $a\alpha + b = c\alpha^2 + d\alpha$ and thus $c\alpha^2 + (d-a)\alpha - b = 0$. If $c = 0$ then we see that, as $\alpha \in \mathbb{R}$ is a fixed point of $\begin{pmatrix} \pm 1 & b \\ 0 & \pm 1 \end{pmatrix}$ that $\alpha = \alpha + b$ whence $b = 0$, contradicting our assumptions on T. Thus $c \neq 0$, whence α is a root of a quadratic polynomial with integer coefficients, as desired. $\square$

§4.3.iv Periodic Continued Fractions

Definition 4.3.26. The continued fraction expansion $[a_0; a_1, a_2, \ldots]$ of an irrational number α is **eventually periodic** if there exist positive integers N and d such that $a_n = a_{n+d}$ for all $n \geq N$. It is periodic if there exists a positive integer d such that $a_n = a_{n+d}$ for all n.

Definition 4.3.27. An irrational number α is a **quadratic irrational** if there is a polynomial $ax^2 + bx + c$, with a, b, c integers, that has α as a root.

Theorem 4.3.28

α has an eventually periodic continued fraction expansion if and only if α is a quadratic irrational.

Proof. ($\implies$): We first show this when α is periodic. We then have a d such that

$$\alpha = a_0 + \cfrac{1}{a_1 + \frac{1}{a_2 + \cdots + \frac{1}{a_{d-1} + \frac{1}{\alpha}}}}.$$

Simplifying the right hand side, we find integers x, y, z, w such that

$$\alpha = \frac{x\alpha + y}{z\alpha + w}.$$

Then $z\alpha^2 + (w - x)\alpha - y = 0$; since α is irrational z cannot be zero, so α is a quadratic irrational.

If α is only eventually periodic there is a β with periodic continued fraction expansion such that we have:

$$\alpha = a_0 + \cfrac{1}{a_1 + \frac{1}{a_2 + \cdots + \frac{1}{a_{N-1} + \frac{1}{\beta}}}}$$

Then β is quadratic irrational. It thus suffices to show that if β is quadratic irrational, so is $\frac{1}{\beta}$ and $n + \beta$ for integer n. Note that if β is a root of $ax^2 + bx + c$, then $\frac{1}{\beta}$ is a root of $a + bx + cx^2$, and $n + \beta$ is a root of $a(x-n)^2 + b(x-n) + c$, so these claims are clear.

($\impliedby$): Fix a real, quadratic irrational α. Then there exist a, b, c integers such that $a\alpha^2 + b\alpha + c = 0$. Let $\alpha = [a_0; a_1, a_2, \ldots]$ be the continued fraction expansion of α, and set $s_n = [a_{n+1}; a_{n+2}, \ldots]$; note that $s_n = \frac{1}{r_n}$, where r_n is the nth remainder. To show that the continued fraction expansion is eventually periodic it suffices to show we get the same remainder twice (that is, $s_i = s_j$ for some $i \neq j$, since the continued fraction expansion of α past s_n depends only on s_n. Moreover, to show that some s_i occurs twice it suffices to show that the set of distinct remainders $\{s_n : n \geq 1\}$ is finite (as then at least one of them must occur infinitely many times). We do this by constructing a finite set of quadratic polynomials such that each s_n is the root of one such polynomial. Since each such polynomial has at most two roots, this will mean that there are finitely many possible s_n, and thus prove the periodicity.

Recall that we have shown that for any n, $\alpha = \frac{p_n + p_{n-1}r_n}{q_n + q_{n-1}r_n}$, where $\frac{p_n}{q_n}$ is the nth convergent. Rewriting in terms of $s_n = \frac{1}{r_n}$ we have

$$\alpha = \frac{p_n s_n + p_{n-1}}{q_n s_n + q_{n-1}}.$$

If we substitute this into $a\alpha^2 + b\alpha + c = 0$ we get the following expression, after clearing denominators.

$$a(p_n s_n + p_{n-1})^2 + b_n(p_n s_n + p_{n-1})(q_n s_n + q_{n-1}) + c(q_n s_n + q_{n-1})^2 = 0$$

and rewriting this as a quadratic in s_n we find that $A_n s_n^2 + B_n s_n + C_n = 0$, where:

$$\begin{aligned} A_n &= ap_n^2 + bp_n q_n + cq_n^2 \\ B_n &= 2ap_n p_{n-1} + b(p_n q_{n-1} + p_{n-1} q_n) + 2cq_n q_{n-1} \\ C_n &= ap_{n-1}^2 + bp_{n-1} q_{n-1} + cq_{n-1}^2 \end{aligned}$$

Note in particular that $C_n = A_{n-1}$. Moreover, computing the discriminant $B_n^2 - 4A_n C_n$ we find (after some algebra) that

$$B_n^2 - 4A_n C_n = (b^2 - 4ac)(p_n q_{n-1} - q_n p_{n-1})^2 = b^2 - 4ac,$$

since we had shown previously that $p_n q_{n-1} - q_n p_{n-1} = \pm 1$.

The idea is now to bound the A_n from above; that is, to show that for some M, $|A_n| \leq M$ independently of n. If we can do this for $|A_n|$, then it is also true that $|C_n| \leq M$ for all M. There would therefore be only finitely many possibilities for the pair (A_n, C_n) as n varies. But once A_n, C_n are fixed, there are at most two possibilities for B_n, since we know that $B_n^2 - 4A_n C_n = b^2 - 4ac$, with a, b, c fixed. Thus if we can bound $|A_n|$, there are only finitely many quadratic polynomials satisfied by the s_n, and thus (as explained above) only finitely many possible s_n, so the continued fraction must be periodic.

Therefore, it remains to show that $|A_n|$ is bounded above. To do this recall that we have shown that $|\frac{p_n}{q_n} - \alpha| < \frac{1}{q_n q_{n+1}}$. Rewriting this we find:

$$|p_n - q_n \alpha| < \frac{1}{q_{n+1}} < \frac{1}{q_n}.$$

As a result we can write $p_n = \alpha q_n + \frac{\delta}{q_n}$ with $|\delta| < 1$. Substituting into $A_n = ap_n^2 + bp_n q_n + cq_n^2$ we find:

$$A_n^2 = a\alpha^2 q_n^2 + 2a\alpha\delta + \frac{\delta^2}{q_n^2} + b\alpha q_n^2 + b\delta + cq_n^2,$$

which we can rewrite as

$$q_n^2(a\alpha^2 + b\alpha + c) + 2a\alpha\delta + \frac{\delta^2}{q_n^2} + b\delta.$$

Note that the first term here is zero, as α is a root of $ax^2 + bx + c$, the second and fourth terms are bounded independently of n, and the third term is less than 1 in absolute value. Thus the whole expression is bounded independently of n and the result follows. □

§4.4 Continued Fractions and Pell's

Now that we know that continued fraction expansions give good rational approximations, it's natural to ask if there is a relationship between the convergents $\frac{p_n}{q_n}$ to the continued fraction expansion of $\sqrt{d}$ and solutions to Pell's equation:

$$x^2 - dy^2 = 1.$$

In fact, we have the following theorem:

Theorem 4.4.1

Let (X, Y) be a solution to $x^2 - dy^2 = 1$, with $X, Y > 0$. Then there exists an integer n such that $(X, Y) = (p_n, q_n)$, where $\frac{p_n}{q_n}$ is the nth convergent to the continued fraction expansion of $\sqrt{d}$.

Note that the converse is false: not every convergent $\frac{p_n}{q_n}$ gives rise to a solution to Pell's equation. Nonetheless, this gives a much more efficient way to find solutions to Pell's equation than the *try increasing values of Y until one works* approach we were using previously. In particular, (since the q_n are an increasing sequence) the first $\frac{p_n}{q_n}$ in the continued fraction expansion of $\sqrt{d}$ such that $p_n^2 - dq_n^2 = 1$ gives the fundamental solution to $x^2 - dy^2 = 1$.

The key to proving this is to first establish a strengthening of our results on good rational approximations. We already know that the best possible rational approximations are convergents; the following theorem gives us a very strong criterion for detecting these convergents:

Theorem 4.4.2

Let α be a real quadratic irrational and let $\frac{p_n}{q_n}$ be the convergents to its continued fraction expansion. Let p, q be integers with q positive and $q < q_{n+1}$. Suppose that $|p - q\alpha| \leq |p_n - q_n\alpha|$. Then $\frac{p}{q} = \frac{p_n}{q_n}$.

Proof. Let $x = pq_n - qp_n$, and $y = pq_{n+1} - qp_{n+1}$. Then we have:

$$\begin{aligned} p_{n+1}x - p_n y &= pp_{n+1}q_n - pp_nq_{n+1} = (-1)^n p \\ q_{n+1}x - q_n y &= -qp_nq_{n+1} + qp_{n+1}q_n = (-1)^n q \end{aligned}$$

(using $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$.) Note that since q_n, q_{n+1} are positive and $q < q_{n+1}$, this is only possible if x and y have the same sign (consider the absolute values). Using these expressions we can write:

$$q\alpha - p = (-1)^n[q_{n+1}x - q_ny]\alpha - (-1)^n[p_nx - p_ny]$$
$$= (-1)^n[(q_{n+1}\alpha - p_{n+1})x - (q_n\alpha - p_n)y]$$

Recall that we have shown that as n increases, the sign of $\alpha - \frac{p_n}{q_n}$ alternates. Thus $q_{n+1}\alpha - p_{n+1}$ has sign opposite to $q_n\alpha - p_n$. Since x and y have the same sign, we can take absolute values in the above equality and obtain

$$|q\alpha - p| = |x||q_{n+1}\alpha - p_{n+1}| + |y||q_n\alpha - p_n|.$$

Since we have assumed that $|q\alpha - p| \le |q_n\alpha - p_n|$, and x, y are integers, we must have either $y = 0$ or $|y| = 1$, $x = 0$. But if $y = 0$ we have $\frac{p}{q} = \frac{p_{n+1}}{q_{n+1}}$ which is impossible since $\frac{p_{n+1}}{q_{n+1}}$ is in lowest terms and $q < q_{n+1}$. So we must have $|y| = 1$, $x = 0$, but this implies that $\frac{p}{q} = \frac{p_n}{q_n}$. □

Corollary 4.4.3

Let α be a real quadratic irrational and p, q positive integers such that $|\frac{p}{q} - \alpha| < \frac{1}{2q^2}$. Then $\frac{p}{q} = \frac{p_n}{q_n}$ for some n.

Proof. Since the q_n are increasing, we can find a unique n such that $q_n \le q < q_{n+1}$. Suppose $|p - q\alpha| \le |p_n - q_n\alpha|$. Then the previous theorem shows immediately that $\frac{p}{q} = \frac{p_n}{q_n}$. So we may assume that $|p - q\alpha| > |p_n - q_n\alpha|$. Then if $\frac{p}{q} \ne \frac{p_n}{q_n}$, we have

$$\frac{1}{qq_n} \le |\frac{p}{q} - \frac{p_n}{q_n}| \le |\frac{p}{q} - \alpha| + |\frac{p_n}{q_n} - \alpha|.$$

By assumption, $|\frac{p}{q} - \alpha| < \frac{1}{2q^2}$. Moreover, we know that $|\frac{p_n}{q_n} - \alpha| = \frac{1}{q_n}|p_n - q_n\alpha|$ and $|p_n - q_n\alpha| < |p - q\alpha| < \frac{1}{2q}$. Thus $|\frac{p_n}{q_n} - \alpha| < \frac{1}{2qq_n}$. In particular we have

$$\frac{1}{qq_n} \le |\frac{p}{q} - \alpha| + |\frac{p_n}{q_n} - \alpha| < \frac{1}{2q^2} + \frac{1}{2qq_n} = \frac{q + q_n}{2q^2q_n}.$$

We deduce that

$$\frac{q + q_n}{2q^2q_n} \ge \frac{1}{qq_n}.$$

Clearing denominators, we find that $q + q_n > 2q$, implying that $q_n \ge q$. Since we chose n such that $q_n \le q$ we must have $q_n = q$; then p must equal p_n and we are done. □

§4.4.i Returning to Pell's

We now return to Pell's equation. Fix a nonsquare positive integer d. We then have:

Theorem 4.4.4

If p, q satisfy $p^2 - dq^2 = 1$, then $\frac{p}{q}$ is a convergent to $\sqrt{d}$.

Proof. It suffices to show that $|\sqrt{d} - \frac{p}{q}| < \frac{1}{2q^2}$, by the corollary. We have

$$|\sqrt{d} - \frac{p}{q}| = |\frac{q\sqrt{d} - p}{q}| = |\frac{q^2d - p^2}{q(q\sqrt{d} + p)}| = |\frac{1}{q^2(\sqrt{d} - \frac{p}{q})}|.$$

However, $\sqrt{d} > 1$ and $\frac{p}{q} > q$ since $p = \sqrt{1 + dq^2} > q$, so the result follows. □

Corollary 4.4.5

If (x, y) is the fundamental solution of $x^2 - dy^2 = 1$, then $(x, y) = (p_n, q_n)$ for the first n such that $p_n^2 - dq_n^2 = 1$.

5 Primes in Arithmetic Progression

We now ask the following question: given a positive integer n, and an integer a with $(a, n) = 1$, are there infinitely many primes congruent to a mod n?

§5.1 First Cases

We first know that there are infinitely many primes.

Theorem 5.1.1

There are infinitely many primes.

Theorem 5.1.2

There are infinitely many primes congruent to 3 mod 4.

Proof. Let S be the set of primes congruent to 3 modulo 4, and suppose this is finite. Then, consider

$$Q = 2 + \prod_{p \in S} p^2.$$

In mod 4, Q is congruent to 3 mod 4, so Q is divisible by at least one prime congruent to 3 mod 4. On the other hand, Q is not divisible by any primes in S. □

A very similar argument works to show there are infinitely many primes congruent to 5 mod 6. Handling cases beyond that requires new ideas. For instance:

Lemma 5.1.3

Let x be an even integer and let p be a prime dividing $x^2 + 1$. Then p is congruent to 1 mod 4.

Proof. On the one hand p is clearly odd. On the other hand, $x^2 \equiv -1 \pmod{p}$, so -1 is a quadratic residue mod p. Hence p is 1 mod 4. □

Theorem 5.1.4

There are infinitely many primes congruent to 1 mod 4.

Proof. Let S be a finite set of primes congruent to 1 mod 4, and set $Q = 1 + \prod_{p \in S} p^2$. By the lemma, every prime dividing Q is congruent to 1 mod 4, but no prime in S divides Q. □

This suggests a strategy for proving there are infinitely many primes congruent to a mod n for some pair a, n: if we can find a polynomial P such that for any integer

x, every prime dividing $P(nx)$ is congruent to a mod n, then we can try to mimic the proof that there are infinitely many primes congruent to 1 mod 4 to show that there are infinitely many primes congruent to a mod n. When $a = 1$ it turns out this is possible. Indeed, in the following lecture we will show:

Theorem 5.1.5

For any positive integer n, there are infinitely many primes congruent to 1 mod n.

To prove this, we need to understand the notion of cyclotomic polynomials.

§5.2 Cyclotomic Polynomials

Definition 5.2.1. The nth cyclotomic polynomial Φ_n is the product:

$$\Phi_n = \prod_{\substack{1 \le a < n \\ (a,n)=1}} (x - e^{\frac{2\pi i a}{n}})$$

In other words, Φ_n is the monic polynomial whose roots are the primitive nth roots of unity, all with multiplicity one. As a priori, Φ_n is a polynomial with complex coefficients. However, we observe:

Lemma 5.2.2

For any n, we have:

$$x^n - 1 = \prod_{d|n} \Phi_d.$$

Proof. The roots of $x^n - 1$ are the n roots of unity, each with multiplicity one. In other words, they are the primitive dth roots of unity for d dividing n, each with multiplicity one. These are the same as the roots of the product of the Φ_d. □

From this, we deduce

Lemma 5.2.3

For any n, the polynomial Φ_n has integer coefficients.

Proof. We prove this by strong induction on n; the case $n = 1$ is clear. Suppose that Φ_d has integer coefficients for all $d < n$. We have $x^n - 1 = \Phi_n P(x)$, where $P(x)$ is the product of Φ_d for d dividing n and strictly less than n. In particular P is monic with integer coefficients by the induction hypothesis. Let e be the degree of p.

Write $\Phi_n(x) = \sum a_n x_n$ and $P(x) = \sum b_n x_n$. Suppose Φ_n does not have integer coefficients, and let q be the largest integer such that a_q is not an integer. Since P(x) is monic, the coefficient of x^{q+e} in $P(x)\Phi_n$ is $a_q + a_{q+1}b_{e-1} + \cdots + a_{q+e}b_0$; every term except a_q is an integer, so this coefficient is not an integer. Since $P(x)\Phi_n(x) = x^n - 1$ this is impossible. □

In light of this, we can consider Φ_n as a polynomial mod p for various p. In fact, we will show that if p does not divide n, then Φ_n has no repeated roots mod p. In order to do this, we need the notion of the derivative of a polynomial with coefficients in an arbitrary field.

Definition 5.2.4. Let F be a field, and $P = \sum a_n x^n$ a polynomial with coefficients in F. The **derivative** P' of P is the polynomial $\sum_{n=1}^{d} n a_n x^{n-1}$.

Remark 5.2.5 — Note that $(P+Q)' = P' + Q'$ and $(PQ)' = P'Q + Q'P$, as one easily checks.

Lemma 5.2.6

Suppose P has a double root α in F. Then $(x\alpha)$ divides both P and P'.

Proof. Write $P = (x-\alpha)^2 Q$. Then $P' = (x-\alpha)^2 Q' + 2(x-\alpha)Q$. □

Corollary 5.2.7

If $p \nmid n$, then $\Phi_n(x)$ has no repeated roots mod p.

Proof. Since $\Phi_n(x)$ divides $x^n - 1$, it suffices to show that $x^n - 1$ has no repeated roots mod p. But the derivative of $x^n - 1$ is a nonzero multiple of $x^n - 1$ (since p does not divide n) and thus has no common factors with x^{n-1}. □

Theorem 5.2.8

Let p be a prime not dividing n, and let a be an integer. Then p divides $\Phi_n(a)$ if, and only if, a has exact order n mod p.

Proof. Suppose that a has exact order n mod p. Then a is a root of $x^n - 1$ mod p, but is not a root of $x^d - 1$ mod p for any d dividing n. Since $\Phi_d(x)$ divides $x^d - 1$, it follows that a can't be a root of $\Phi_d(x)$ for any d dividing n other than $\Phi_n(x)$. Since $a^n - 1 = \prod_{d|n} \Phi_d(a)$, we must have $\Phi_n(a) \equiv 0 \pmod p$.

Conversely, suppose that $\Phi_n(a) \equiv 0 \pmod p$. Then $a^n - 1$ is zero mod p, so the order of a mod p divides n. Let d be this order, and suppose $d < n$. We have $a^d - 1 \equiv 0 \pmod p$ which means that $\Phi_e(a)$ is zero for some $e \mid d$. On the other hand then a is a root of $\Phi_n(x)$ and $\Phi_e(x)$, which are distinct factors of $x^n - 1$. Thus a is a double root of $x^n - 1$ which is impossible by the lemma. Therefore $d = n$ and we are done. □

Corollary 5.2.9

Let p be a prime not dividing n, and a an integer. If p divides $\Phi_n(a)$, then $p \equiv 1 \pmod n$.

Proof. The order of a mod p is n by the theorem above. Thus n divides $p-1$ by Fermat's little theorem. □

We're now ready to prove the goal:

Theorem 5.2.10

Let n be a positive integer. There are infinitely many primes congruent to 1 mod n.

Proof. Let S be a finite set of primes congruent to 1 mod n. Let R be the product of the primes in S, and for each k, let Q_k be the integer $\Phi_n(knR)$. Note that not all Q_k are ± 1, since Φ_n is a nonconstant polynomial. Thus, for some k, there is a prime p dividing Q_k. Since Q_k divides $(knR)^n - 1$, no prime dividing n or R can divide Q_k. Thus p is not in S, and by the corollary p is congruent to 1 mod n. $\square$